

Submersible Motor Pump

Amarex

Sizes DN 50 to DN 150

Motors:

2 Poles: 014 to 084

4 Poles: 012 to 077

Type Series Booklet



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Type Series Booklet Amarex

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Waste Water

Submersible Motor Pump

Amarex



Main applications

- Waste water transport
- Waste water management
- Drainage systems
- Waste water treatment plants
- Stormwater transport
- Recirculation
- Sludge treatment

Fluids handled

- Service water
- Grey water
- Waste water with faeces
- Waste water containing long fibres and solid substances
- Fluids containing gas
- Activated sludge
- Digested sludge
- Raw sludge

Further information on fluids handled

Overview of fluids handled (⇒ Page 12)

Operating data

Operating properties

Characteristic		Value
Flow rate	Q [m³/h]	≤ 320
	Q [l/s]	≤ 89
Head	H [m]	≤ 42
Fluid temperature	T [°C]	≤ +40
Motor rating	P ₂ [kW]	0,35 - 8,4

Design details

Design

- Fully floodable submersible motor pump
- Not self-priming
- Close-coupled design

Drive

- Three-phase asynchronous squirrel-cage motor
- Type of protection Ex db IIB (applies to explosion-proof pump sets only)
- Enclosure: IP68 to EN 60529/IEC529

Shaft seal

- Two bi-directional mechanical seals in tandem arrangement, with liquid reservoir

Impeller type

- Various application-oriented impeller types

Bearings

Motor-end bearings:

- Grease-packed bearings sealed for life
- Maintenance-free

Pump-end bearings:

- Grease-packed bearings sealed for life
- Maintenance-free
- Reinforced bearings¹⁾

1) Standard for impeller type D-max, optional for impeller type F-max

Designation

Designation example

Position																																				
1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37
A	R	X		D	1	0	0	-	2	3	0	/	0	3	8	C	4	U	S	G	-	-	2	2	0	-	0	0	0	0	0	X	1	0	0	S
See name plate and data sheet																										See data sheet										

Designation key

Position	Code	Description
1-3	Pump type	
	ARX	Amarex
5	Impeller type	
	D-max	Open two-vane impeller
	F-max	Free-flow impeller
6-12	Size	
	100	Nominal discharge nozzle diameter [mm]
	230	Hydraulic system size
14-16	Motor rating P _N [kW]	
	038	3,80
	...	...
	108	10,80
17	Efficiency class ²⁾	
	C	IE3
	D	IE2
	E	IE1
	F	None
18	Number of motor poles	
	2	2 poles
	4	4 poles
19	Motor version	
	U	Without explosion protection, standard
	Y	Explosion-proof
20	Motor construction type	
	S	Wet installation
21-22	Material variant	
	G	Standard variant grey cast iron ³⁾
	G1	Standard variant grey cast iron, impeller made of duplex stainless steel
	G2	Standard variant grey cast iron, impeller made of white cast iron
	GH	Standard variant grey cast iron, impeller and discharge cover made of white cast iron
24-26	Nominal impeller diameter [mm]	
	090	90
	...	...
	220	220
28	Type of power cable	
	0	H07RN_F (standard)
	1	S1BN8_F (standard)
	2	TEHSITE (resistant to chemicals)
	3	S07RC4N8_F (shielded)
29	Type of cable gland	
	0	CUZN40PB2 (standard)
	1	CW617N (resin)
	2	To manufacturer's choice (Gisma)
30	Leakage sensors, drive end	

- 2) IEC 60034-30 standard not binding for submersible motor pumps. Efficiencies calculated/determined according to the measurement method specified in IEC 60034-2. The marking is used for submersible motors that achieve efficiency levels similar to those of standardised motors acc. to the IEC 60034-30 standard.
- 3) Depending on the configuration the impeller and suction cover of versions with D-max impeller are made of nodular cast iron.

Position	Code	Description
30	1	None
	2	Leakage sensor
31	Explosion protection standard	
	0	None
	1	ATEX
	2	CEC (FM)
	3	CSA
	4	EACEx
32	Oil fill of the motor	
	0	S (without oil fill)
33	Motor size	
	S	NG08M
	M	NG09M
	L	NG10M
	X	NG10M/NG10L
34	Basic symbol roller bearing, pump end	
	0	6306 (single-row roller bearing)
	1	3306 (double-row roller bearing)
35	Shaft seal material, pump end	
	0	Q1Q1PGG
	1	Q1Q1VGG
	2	Q12Q1VGG1
	3	Q12Q1KGG1
36	D-flector	
	0	Without D-flector in the suction cover
	1	With D-flector in the suction cover
37	00000X100S	Additional code for design variant

Materials

Overview of available materials

Part No. (⇒ Page 70)	Description		Version	Material variant			
				G	G1 ⁴⁾	G2	GH
Pump set							
100	Casing			EN GJL-250			
160	Cover (motor)			EN GJL-250			
162	Suction cover		Standard	EN GJL-250			EN GJL-250 ⁴⁾
			With D-flector ⁵⁾	EN GJS-600-3	-	EN GJS-600-3	EN-GJN-HB555
163	Discharge cover			EN GJL-250			EN-GJN-HB555
210	Shaft			1.4021			
230	Impeller		F-max	EN GJL-250	1.4517	EN-GJN-HB555	EN-GJN-HB555
			D-max	EN GJS-600-3	-	EN-GJN-HB555	EN-GJN-HB555
412	O-ring		Standard	Nitrile butadiene rubber (NBR70)			
			Option	Fluoroelastomer (FKM80)			
				Viton, Teflon-coated (FEP/FKM)			
433	Mechanical seal	Drive end	Standard	Carbon/Al2O3			
		Pump end	Standard	SIC/SIC/NBR			
			Option	SIC/SIC/FPM			
				SIC/SIC/KAL			
914	Hexagon socket head cap screw			A4			
Installation parts							
182	Foot			1.4301			
572	Guide wire suspension bracket			1.4571			
59-24	Wire			1.4401			
72-1	Flanged bend			EN GJL-250			
732	Claw			EN GJL-250			
885	Lifting chain / lifting rope			Lifting chain: 1.4404			
			Lifting rope: polypropylene				
892	Foot plate			1.4301			
894	Mounting bracket			1.4301			

Description of materials

Grey cast iron EN-GJL-250 (lamellar graphite cast iron)

Lamellar graphite cast iron to EN 1561 is the most widely used cast material for handling municipal sewage, waste water and sludges as well as stormwater and surface water. It is suitable for neutral fluids which are only slightly aggressive and cause little wear. The pH should be ≥ 6.5 , the sand content ≤ 0.5 g/l.

Duplex stainless steel (1.4517 or technically equivalent material)

This type of carbon steel is resistant to cavitation, has excellent strength values and is used for high circumferential speeds. An excellent resistance to pitting corrosion makes ferritic-austenitic stainless carbon steel a popular choice for pumping acidic waste water with a high chloride content as well as seawater and brackish water. Thanks to its good chemical resistance, e.g. against waste water containing phosphorous and sulphuric acid, this material is used in a wide range of applications in the chemical industry and process engineering. Pumps made of duplex stainless steel have a very long service life, even when handling brines, chemical waste water (pH 1 - 12), grey water and landfill leachate.

Wear-resistant white cast iron (EN-GJN-HB555 [XCR14] or technically equivalent material)

Wear-resistant white cast iron is suitable for handling highly abrasive fluids containing sand, ash or iron ore sinter, for example. It has a Rockwell hardness of at least 54 HRC, which is higher than that of hardened chrome steel. Owing to its hardness, the chromium-molybdenum alloy cast iron features a notably higher wear resistance than EN-GJL-250 grey cast iron and other cast materials. The pH should be ≥ 6.5 .

4) Only available for version with F-max impeller
5) D-flector only available for version with D-max impeller

Product benefits

- High efficiency thanks to enhanced power density (Wh/m³), overall efficiency of up to 69 %, lower total cost of ownership
- Non-clogging (no clogging by wet wipes) with patented deflection technology and D-max impeller
- Long service life with high operating reliability and durable bearings (100,000 operating hours)
- Environmentally friendly with non-toxic, food-approved oil fill for lubricating the mechanical seal (liquid reservoir)
- Flexible thanks to easy substitution of competitor pumps with a large range of adapter claws and substitution pump app

Product information

Product information as per Regulation No. 1907/2006 (REACH)

For information as per chemicals Regulation (EC) No. 1907/2006 (REACH), see <http://www.ksb.com/reach>.

Overview of product features / selection tables
Overview of product features, material variant G

Overview of product features, material variant G

		Motor sizes			
		23F 2	24F 2 ... 40F 2	51F 2 ... 73F 2	84F 2
		14C 2	18C 2 ... 29C 2	35C 2 ... 45C 2	60C 2
		-	17F 4 ... 23F 4	35F 4 ... 42F 4	65F 4 ... 77F 4
		-	-	21C 4	36C 4 ... 45C 4
Voltage		230 V / 380 V / 400 V / 415 V / 500 V / 690 V			
Starting method		DOL		DOL / star-delta	
Cooling		Cooled by surrounding fluid			
Duty cycle		S1 - submerged, max. 25 m (see ²⁾ in outline drawing) S1 - outside the fluid, minimum water level R3 and power derating to IE3			
Frequency of starts/hour		30		30 (P ₂ ≤ 7,5 kW) 10 (P ₂ > 7,5 kW)	
Bearings					
Bearing design	D-max	-	Double-row angular contact ball bearing		
	F-max	Deep groove ball bearing			
Lubrication		Grease-packed rolling element bearings sealed for life			
Sealing element					
Monitoring					
Winding temperature motor version US		Temperature monitoring circuit (with automatic reset and start-up): bimetal switch directly connected with the control circuit of the motor contactor			
Winding temperature motor version YS		Limiting circuit (temperature limit for explosion protection without automatic reset): bimetal switch connected via a tripping unit with manual reset			
Power cable					
Type		Rubber-sheathed cable (H07RN8-F)			
Length	Motor version US	8 m			
	Motor version YS	10 m			
Cable entry		Cable gland			
Sealing elements					
Shaft seal		Mechanical seal (bellows-type mechanical seal) (Q1Q1PGG)			
Elastomers		NBR			
Explosion protection					
Motor version US		Not explosion-proof			
Motor version YS		Ⓔ II 2G Ex db h IIB T4 Gb			
Coating		Environmentally friendly KSB top coat (two-component epoxy paint), colour RAL 5002, film thickness = 80 µm			
Maximum fluid temperature		40 °C			
Installation (⇒ Page 53)					
Stationary, with guide hoop	F-max	Installation depths 1.5 m/1.8 m/2.1 m			-
Stationary, with single guide rail		Installation depth: 6 m	-	-	
Stationary, with twin guide rail		Installation depth: 12 m			
Stationary, with guide wire		Installation depth: 4.5 m			
Transportable		Installation depth: 14.5 m			

6) 690 V DOL starting only
7) 10 m / 15 m / 20 m / 30 m / 40 m / 50 m available

Overview of product features, material variants G1, G2, GH

Overview of product features, material variants G1, G2, GH

		Motor sizes			
		23F 2	24F 2 ... 40F 2	51F 2 ... 73F 2	84F 2
		14C 2	18C 2 ... 29C 2	35C 2 ... 45C 2	60C 2
		-	17F 4 ... 23F 4	35F 4 ... 42F 4	65F 4 ... 77F 4
		-	-	21C 4	36C 4 ... 45C 4
Voltage		230 V / 380 V / 415 V / 500 V / 690 V ⁶⁾			
Starting method		DOL		DOL / star-delta	
Bearings					
Bearing design		F-max	Double-row angular contact ball bearing		
Power cable					
Type		Rubber-sheathed cable (H07RN8-F) Shielded connection cable (S07RC4N8-F) TEFZEL			
Length		≤ 50 m ⁷⁾			
Cable entry		Water-tight cable entry			
Monitoring					
Leakage inside the motor		Leakage sensor in the motor			
Sealing elements					
Shaft seal		Mechanical seal (bellows-type mechanical seal) Q1Q1PGG Mechanical seal with covered springs			
Material variant, shaft seal (pump end)		Q1Q1VGG Q12Q1VGG1 Q12Q1KGG1			
Elastomers		FKM FKM / Teflon			
Coating		Environmentally friendly KSB top coat (two-component epoxy paint), colour RAL 5002, film thickness = 80 µm			
Installation (⇒ Page 53)					
Stationary, with twin guide rail		Installation depth: 12 m			
Stationary, with guide wire		Installation depth: 14.5 m			

Impellers

	Free-flow impeller (impeller type F-max)	Suitable for the following fluids: fluids containing solids and stringy material as well as fluids with entrapped air or entrapped gas
	Open two-vane impeller (impeller type D-max)	Suitable for the following fluids: waste water containing wet wipes and long fibres

Overview of fluids handled

KSB EasySelect, selection software for all applications



KSB EasySelect is a comprehensive selection tool for all applications. It guides users to an optimal solution for their projects by offering a fast, easy and user-friendly way to select and configure pumps and valves. All that is required are some project-specific criteria and a few minutes' time. The tool systematically guides the user through KSB's wide range of products to the right product for the application at hand.



Other fluids upon request.

The table below for your guidance is based on KSB's long-standing experience. The data are standard values and are not to be considered as generally binding recommendations. More detailed advice is available from our specialist department. Make use of our laboratory's expertise when selecting materials.

Selection aid for materials and hydraulic systems per fluid

Fluid handled ⁸⁾	Recommend - ed material	Recommend - ed impeller type ⁹⁾	Recommend - ed sealing elements	Comments, further recommendations
Water, surface water				
▪ Dam water	G	F-max, D-max	NBR	Free passage > any solids contained, possibly pre-screened
▪ Lake water	G	F-max, D-max	NBR	Free passage > any solids contained, possibly pre-screened
▪ River water	G	F-max, D-max	NBR	Free passage > any solids contained, possibly pre-screened
Water, contaminated water				
▪ Mixed water, with strainer	G	F-max, D-max	NBR	-
▪ Mixed water, without strainer	G	F-max, D-max	NBR	-
▪ Slightly contaminated water	G	F-max, D-max	NBR	Free passage > any solids contained, possibly pre-screened
▪ Waste water with faeces	G	D-max, F-max	NBR	EN 12050, min. free passage of 40 mm
▪ Waste water without faeces	G	F-max, D-max	NBR	-
Municipal waste water				
▪ Biologically treated	G	F-max, D-max	NBR	-
▪ Containing air and gas	G	F-max	NBR	Up to 8 %; contact KSB for higher concentrations.
▪ Domestic waste water with faeces	G	F-max, D-max	NBR	EN 12050, min. free passage of 40 mm
▪ Domestic waste water without faeces	G	F-max, D-max	NBR	-
▪ Pumped drainage	G	F-max, D-max	NBR	-
▪ Raw waste water containing solids, long fibres and abrasive particles	G	F-max, D-max	NBR	Free passage > any solids contained, possibly pre-screened
▪ Untreated	G	F-max, D-max	NBR	ATV ¹⁰⁾ recommends a free passage of 100 mm; min. free passage: 76 mm
Abrasive municipal waste water				
▪ Raw waste water containing solids, long fibres and abrasive particles	GH	D-max, F-max		Solids content < 5 g/l
Water, raw water				
▪ No details specified	G	F-max, D-max	NBR	-
Corrosive and abrasive industrial waste water				
▪ No details specified	G1	F-max	NBR	-
Abrasive, non-corrosive industrial waste water¹¹⁾				
▪ Containing dust / ash	G2, GH	F-max, D-max	NBR	-
▪ Containing lime milk > 5 to 15 %	G2, GH	F-max, D-max	NBR	-

8) For any fluids which are not listed in this table contact KSB.

9) The first impeller type listed should be given preference.

10) ATV = German regulatory body for waste water management

11) Severe hydroabrasive wear occurs if solids contents of approx. 0.5 g/l or higher are combined with circumferential speeds exceeding 20 m/s or part load conditions to the left of the duty point. The material variants required highly depend on the operating hours, rotational speed and flow velocity.

Fluid handled ⁸⁾	Recommend - ed material	Recommend - ed impeller type ⁹⁾	Recommend - ed sealing elements	Comments, further recommendations
▪ Pigment suspensions	G2, GH	F-max, D-max	NBR	-
▪ Lime water	G2, GH	F-max, D-max	NBR	-
▪ Wash water containing solids	G2, GH	F-max, D-max	NBR	-
Non-abrasive, non-corrosive industrial waste water¹²⁾				
▪ Industrial waste water with faeces	G	F-max, D-max	NBR	-
▪ Industrial waste water without faeces	G	F-max, D-max	NBR	-
▪ Containing aliphatic hydrocarbons	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Ammonium hydroxide	G	F-max	NBR	-
▪ Containing up to 5 % of ammonium hydroxide	G	F-max	NBR	-
▪ Containing aromatic hydrocarbons	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing benzene	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing 5 % of calcium hydroxide Ca(OH) ₂	G	F-max	NBR	-
▪ Containing chlorinated hydrocarbons	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing chloroform	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing 10 % of dissolved carbonate Na ₂ CO ₃	G	F-max	NBR	-
▪ Containing ethylene chloride	G	F-max	FEP-FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing fibres	G	D-max, F-max	NBR	-
▪ Containing methane	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing methylene chloride	G	F-max	FEP-FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing oil	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing paint suspension	G	F-max	NBR	Solvent-free, observe the operator's instructions.
▪ Containing petrol	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing 10 % of potassium hydroxide KOH	G	F-max	FEP-FKM	-
▪ Containing 5 % of sodium hydroxide NaOH	G	F-max	FEP-FKM	-
▪ Containing styrene	G	F-max	FEP-FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing tetrachloroethylene	G	F-max	FKM	TEHSITE cable; for high concentrations contact KSB.
▪ Containing 25 % of urea (NH ₂) ₂ -CO	G	F-max	NBR	-
Acidic water				
▪ Slightly acidic, pH ≥ 6	G1	F-max	FKM	pH ≥ 6 G1 variant and FPM (Viton) O-rings
Suspensions containing solids				
▪ Pulp, concentration up to 1 % bone dry	G	F-max, D-max	NBR	-
▪ Water/sand mixture up to 5 g/l	G2, GH	F-max, D-max	NBR	-
▪ Pulp, concentration up to 6 % bone dry	G	F-max, D-max	NBR	-
▪ Water/sand mixture up to 500 g/l	G	F-max, D-max	NBR	-
Sludges				

12) The hydrocarbons mentioned may occur in very high concentrations due to the difference in specific weight and their low solubility. If this is the case, contact KSB.

Fluid handled ⁸⁾	Recommend - ed material	Recommend - ed impeller type ⁹⁾	Recommend - ed sealing elements	Comments, further recommendations
▪ Raw sludge	G	D-max, F-max	NBR	Pumpable up to a dry substance content of: 13 % (D-max), 8 % (F-max)
▪ Digested sludge	G	D-max, F-max	NBR	Pumpable up to a dry substance content of: 13 % (D-max), 8 % (F-max)
▪ Activated sludge	G	D-max, F-max	NBR	Pumpable up to a dry substance content of: 13 % (D-max), 8 % (F-max)

Technical data
Amarex G, free-flow impeller, n = 2900 rpm

G = material variant grey cast iron (pump casing) / grey cast iron (intermediate casing) / grey cast iron (impeller)

F = free-flow impeller

US = without explosion protection, for fluid temperatures of up to 40 °C

YS = explosion protection Ex II2G Ex db h IIB T4 Gb, for fluid temperatures of up to 40 °C

Amarex	Impeller diameter	Motor construction type	Motor design	P ₂	P ₁	I _N	I _A	T ¹³⁾	Mat. No.
	[mm]			[kW]	[kW]	[A]	[A]	[°C]	
F050-140/024	100	S	U	2,40	3,01	5,25	45	40	39110001
F050-140/024	100	S	Y	2,40	3,01	5,25	45	40	39110002
F050-140/024	110	S	U	2,40	3,01	5,25	45	40	39110003
F050-140/024	110	S	Y	2,40	3,01	5,25	45	40	39110004
F050-140/024	120	S	U	2,40	3,01	5,25	45	40	39110005
F050-140/024	120	S	Y	2,40	3,01	5,25	45	40	39110006
F050-140/024	130	S	U	2,40	3,01	5,25	45	40	39110007
F050-140/024	130	S	Y	2,40	3,01	5,25	45	40	39110008
F050-140/024	140	S	U	2,40	3,01	5,25	45	40	39110009
F050-140/024	140	S	Y	2,40	3,01	5,25	45	40	39110010
F050-140/024	150	S	U	2,40	3,01	5,25	45	40	39110011
F050-140/024	150	S	Y	2,40	3,01	5,25	45	40	39110012
F050-140/024	160	S	U	2,40	3,01	5,25	45	40	39110013
F050-140/024	160	S	Y	2,40	3,01	5,25	45	40	39110014
F050-220/024	160	M	US	2,40	2,95	4,89	45	40	39110027
F050-220/024	160	M	YS	2,40	2,95	4,89	45	40	39110028
F050-220/045	160	M	U	4,50	4,86	8,17	76	40	39110031
F050-220/045	160	M	Y	4,50	4,86	8,17	76	40	39110032
F050-220/045	170	M	U	4,50	4,86	8,17	76	40	39110033
F050-220/045	170	M	Y	4,50	4,86	8,17	76	40	39110034
F050-220/040	180	M	US	4,00	4,86	8,17	76	40	39110045
F050-220/040	180	M	YS	4,00	4,86	8,17	76	40	39110046
F050-220/045	180	M	U	4,50	6,10	10,24	86	40	39110035
F050-220/045	180	M	Y	4,50	6,10	10,24	86	40	39110036
F050-220/051	170	L	US	5,10	6,10	10,24	86	40	39110037
F050-220/051	170	L	YS	5,10	6,10	10,24	86	40	39110038
F050-220/051	180	L	US	5,10	6,10	10,24	86	40	39110039
F050-220/051	180	L	YS	5,10	6,10	10,24	86	40	39110040
F050-220/073	190	L	US	7,30	8,72	14,65	125	40	39110041
F050-220/073	190	L	YS	7,30	8,72	14,65	125	40	39110042
F065-170/028	110	M	U	2,80	2,95	4,89	45	40	39110077
F065-170/028	110	M	Y	2,80	2,95	4,89	45	40	39110078
F065-170/028	120	M	U	2,80	2,95	4,89	45	40	39110079
F065-170/028	120	M	Y	2,80	2,95	4,89	45	40	39110080
F065-170/028	130	M	U	2,80	2,95	4,89	45	40	39110081
F065-170/028	130	M	Y	2,80	2,95	4,89	45	40	39110082
F065-170/028	140	M	U	2,80	2,95	4,89	45	40	39110083
F065-170/028	140	M	Y	2,80	2,95	4,89	45	40	39110084
F065-170/045	150	M	U	4,50	4,86	8,17	76	40	39110085
F065-170/045	150	M	Y	4,50	4,86	8,17	76	40	39110086
F065-170/045	160	M	U	4,50	4,86	8,17	76	40	39110087
F065-170/045	160	M	Y	4,50	4,86	8,17	76	40	39110088
F065-170/045	170	M	U	4,50	4,86	8,17	76	40	39110089
F065-170/045	170	M	Y	4,50	4,86	8,17	76	40	39110090

13) Fluid temperature

Amarex	Impeller diameter	Motor construction type	Motor design	P ₂	P ₁	I _N	I _A	T ¹³⁾	Mat. No.
	[mm]			[kW]	[kW]	[A]	[A]	[°C]	
F065-170/040	180	M	US	4,00	4,86	8,17	76	40	39110097
F065-170/040	180	M	YS	4,00	4,86	8,17	76	40	39110098
F065-170/051	180	L	US	5,10	6,10	10,24	86	40	39110091
F065-170/051	180	L	YS	5,10	6,10	10,24	86	40	39110092
F065-170/073	190	L	US	7,30	8,72	14,65	125	40	39110093
F065-170/073	190	L	YS	7,30	8,72	14,65	125	40	39110094
F080-220/028	140	M	U	2,80	2,95	4,89	45	40	39110163
F080-220/028	140	M	Y	2,80	2,95	4,89	45	40	39110164
F080-220/045	150	M	U	4,50	4,86	8,17	76	40	39110165
F080-220/045	150	M	Y	4,50	4,86	8,17	76	40	39110166
F080-220/045	160	M	U	4,50	4,86	8,17	76	40	39110167
F080-220/045	160	M	Y	4,50	4,86	8,17	76	40	39110168
F080-220/040	170	M	US	4,00	4,86	8,17	76	40	39110183
F080-220/040	170	M	YS	4,00	4,86	8,17	76	40	39110184
F080-220/051	170	L	US	5,10	6,10	10,24	86	40	39110169
F080-220/051	170	L	YS	5,10	6,10	10,24	86	40	39110170
F080-220/073	180	L	US	7,30	8,72	14,65	125	40	39110171
F080-220/073	180	L	YS	7,30	8,72	14,65	125	40	39110172
F080-220/073	190	L	US	7,30	8,72	14,65	125	40	39110173
F080-220/073	190	L	YS	7,30	8,72	14,65	125	40	39110174
F080-220/084	200	X	US	8,40	9,85	16,78	157	40	39110175
F080-220/084	200	X	YS	8,40	9,85	16,78	157	40	39110176

Amarex G, free-flow impeller, n = 1450 rpm

G = material variant grey cast iron (pump casing) / grey cast iron (intermediate casing) / grey cast iron (impeller)

F = free-flow impeller

US = without explosion protection, for fluid temperatures of up to 40 °C

YS = explosion protection Ex II2G Ex db h IIB T4 Gb, for fluid temperatures of up to 40 °C

Amarex	Impeller diameter	Motor construction type	Motor design	P ₂	P ₁	I _N	I _A	T ¹⁴⁾	Mat. No.
	[mm]			[kW]	[kW]	[A]	[A]	[°C]	
F065-150/020	150	M	U	2,00	2,13	3,71	27	40	39110055
F065-150/020	150	M	Y	2,00	2,13	3,71	27	40	39110056
F065-150/020	160	M	U	2,00	2,13	3,71	27	40	39110057
F065-150/020	160	M	Y	2,00	2,13	3,71	27	40	39110058
F065-150/020	170	M	U	2,00	2,13	3,71	27	40	39110059
F065-150/020	170	M	Y	2,00	2,13	3,71	27	40	39110060
F065-150/020	180	M	U	2,00	2,13	3,71	27	40	39110061
F065-150/020	180	M	Y	2,00	2,13	3,71	27	40	39110062
F065-150/020	190	M	U	2,00	2,13	3,71	27	40	39110063
F065-150/020	190	M	Y	2,00	2,13	3,71	27	40	39110064
F065-150/020	200	M	U	2,00	2,13	3,71	27	40	39110065
F065-150/020	200	M	Y	2,00	2,13	3,71	27	40	39110066
F065-230/020	160	M	U	2,00	2,13	3,71	27	40	39110107
F065-230/020	160	M	Y	2,00	2,13	3,71	27	40	39110108
F065-230/020	170	M	U	2,00	2,13	3,71	27	40	39110109
F065-230/020	170	M	Y	2,00	2,13	3,71	27	40	39110110
F065-230/020	180	M	U	2,00	2,13	3,71	27	40	39110111
F065-230/020	180	M	Y	2,00	2,13	3,71	27	40	39110112
F065-230/020	190	M	U	2,00	2,13	3,71	27	40	39110113
F065-230/020	190	M	Y	2,00	2,13	3,71	27	40	39110114
F065-230/025	200	M	U	2,50	2,98	5,15	38	40	39110115
F065-230/025	200	M	Y	2,50	2,98	5,15	38	40	39110116
F065-230/040	210	L	U	4,00	4,41	7,56	45	40	39110117
F065-230/040	210	L	Y	4,00	4,41	7,56	45	40	39110118
F080-150/020	140	M	U	2,00	2,13	3,71	27	40	39110129
F080-150/020	140	M	Y	2,00	2,13	3,71	27	40	39110130
F080-150/020	160	M	U	2,00	2,13	3,71	27	40	39110131
F080-150/020	160	M	Y	2,00	2,13	3,71	27	40	39110132
F080-180/020	140	M	U	2,00	2,13	3,71	27	40	39110143
F080-180/020	140	M	Y	2,00	2,13	3,71	27	40	39110144
F080-180/020	150	M	U	2,00	2,13	3,71	27	40	39110145
F080-180/020	150	M	Y	2,00	2,13	3,71	27	40	39110146
F080-180/020	160	M	U	2,00	2,13	3,71	27	40	39110147
F080-180/020	160	M	Y	2,00	2,13	3,71	27	40	39110148
F080-180/020	170	M	U	2,00	2,13	3,71	27	40	39110149
F080-180/020	170	M	Y	2,00	2,13	3,71	27	40	39110150
F080-180/020	180	M	U	2,00	2,13	3,71	27	40	39110151
F080-180/020	180	M	Y	2,00	2,13	3,71	27	40	39110152
F080-230/020	150	M	U	2,00	2,13	3,71	27	40	39110193
F080-230/020	150	M	Y	2,00	2,13	3,71	27	40	39110194
F080-230/020	160	M	U	2,00	2,13	3,71	27	40	39110195
F080-230/020	160	M	Y	2,00	2,13	3,71	27	40	39110196
F080-230/025	170	M	U	2,50	2,98	5,15	38	40	39110197
F080-230/025	170	M	Y	2,50	2,98	5,15	38	40	39110198
F080-230/040	180	L	U	4,00	4,41	7,56	45	40	39110199
F080-230/040	180	L	Y	4,00	4,41	7,56	45	40	39110200

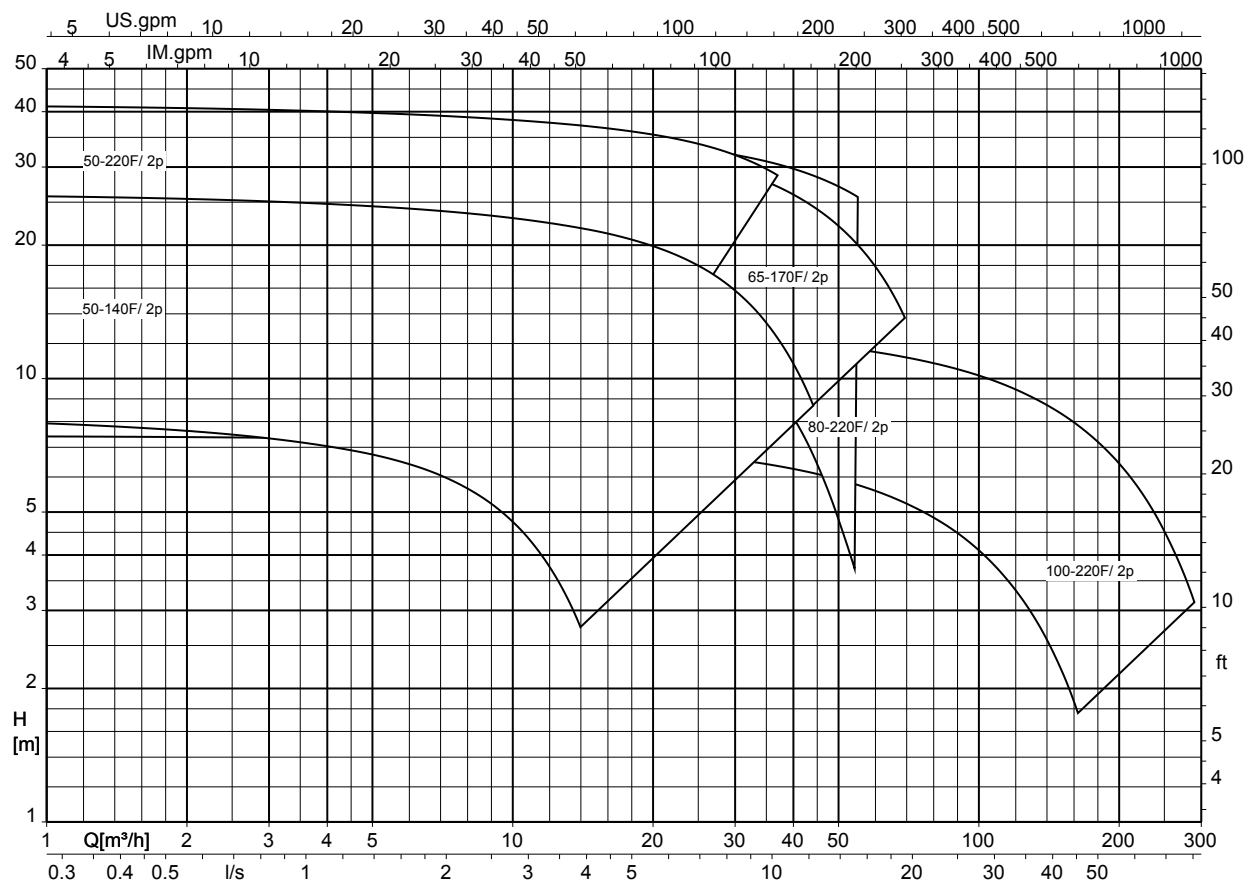
14) Fluid temperature

Amarex	Impeller diameter	Motor construction type	Motor design	P ₂	P ₁	I _N	I _A	T ¹⁴⁾	Mat. No.
	[mm]			[kW]	[kW]	[A]	[A]	[°C]	
F080-230/040	190	L	U	4,00	4,41	7,56	45	40	39110201
F080-230/040	190	L	Y	4,00	4,41	7,56	45	40	39110202
F080-230/055	200	L	U	5,50	5,40	10,46	81	40	39110203
F080-230/055	200	L	Y	5,50	5,40	10,46	81	40	39110204
F080-230/055	210	L	U	5,50	5,40	10,46	81	40	39110205
F080-230/055	210	L	Y	5,50	5,40	10,46	81	40	39110206
F100-180/020	140	M	U	2,00	2,13	3,71	27	40	39110221
F100-180/020	140	M	Y	2,00	2,13	3,71	27	40	39110222
F100-180/020	160	M	U	2,00	2,13	3,71	27	40	39110223
F100-180/020	160	M	Y	2,00	2,13	3,71	27	40	39110224
F100-180/020	180	M	U	2,00	2,13	3,71	27	40	39110225
F100-180/020	180	M	Y	2,00	2,13	3,71	27	40	39110226
F100-180/020	190	M	U	2,00	2,13	3,71	27	40	39110227
F100-180/020	190	M	Y	2,00	2,13	3,71	27	40	39110228
F100-230/020	160	M	U	2,00	2,13	3,71	27	40	39110259
F100-230/020	160	M	Y	2,00	2,13	3,71	27	40	39110260
F100-230/025	170	M	U	2,50	2,98	5,15	38	40	39110261
F100-230/025	170	M	Y	2,50	2,98	5,15	38	40	39110262
F100-230/040	180	L	U	4,00	4,41	7,56	45	40	39110263
F100-230/040	180	L	Y	4,00	4,41	7,56	45	40	39110264
F100-230/040	190	L	U	4,00	4,41	7,56	45	40	39110265
F100-230/040	190	L	Y	4,00	4,41	7,56	45	40	39110266
F100-230/055	200	L	U	5,50	5,40	10,46	81	40	39110267
F100-230/055	200	L	Y	5,50	5,40	10,46	81	40	39110268
F100-230/055	210	L	U	5,50	5,40	10,46	81	40	39110269
F100-230/055	210	L	Y	5,50	5,40	10,46	81	40	39110270
F150-180/065	160	X	US	6,50	7,89	13,35	106	40	39110285
F150-180/065	160	X	YS	6,50	7,89	13,35	106	40	39110286
F150-180/065	170	X	US	6,50	7,89	13,35	106	40	39110287
F150-180/065	170	X	YS	6,50	7,89	13,35	106	40	39110288
F150-180/065	180	X	US	6,50	7,89	13,35	106	40	39110289
F150-180/065	180	X	YS	6,50	7,89	13,35	106	40	39110290
F150-230/065	170	X	US	6,50	7,89	13,35	106	40	39110301
F150-230/065	170	X	YS	6,50	7,89	13,35	106	40	39110302
F150-230/065	180	X	US	6,50	7,89	13,35	106	40	39110303
F150-230/065	180	X	YS	6,50	7,89	13,35	106	40	39110304
F150-230/065	190	X	US	6,50	7,89	13,35	106	40	39110305
F150-230/065	190	X	YS	6,50	7,89	13,35	106	40	39110306
F150-230/065	200	X	US	6,50	7,89	13,35	106	40	39110307
F150-230/065	200	X	YS	6,50	7,89	13,35	106	40	39110308
F150-230/065	210	X	US	6,50	7,89	13,35	106	40	39110309
F150-230/065	210	X	YS	6,50	7,89	13,35	106	40	39110310
F150-230/077	220	X	US	7,70	9,20	15,86	118	40	39110311
F150-230/077	220	X	YS	7,70	9,20	15,86	118	40	39110312

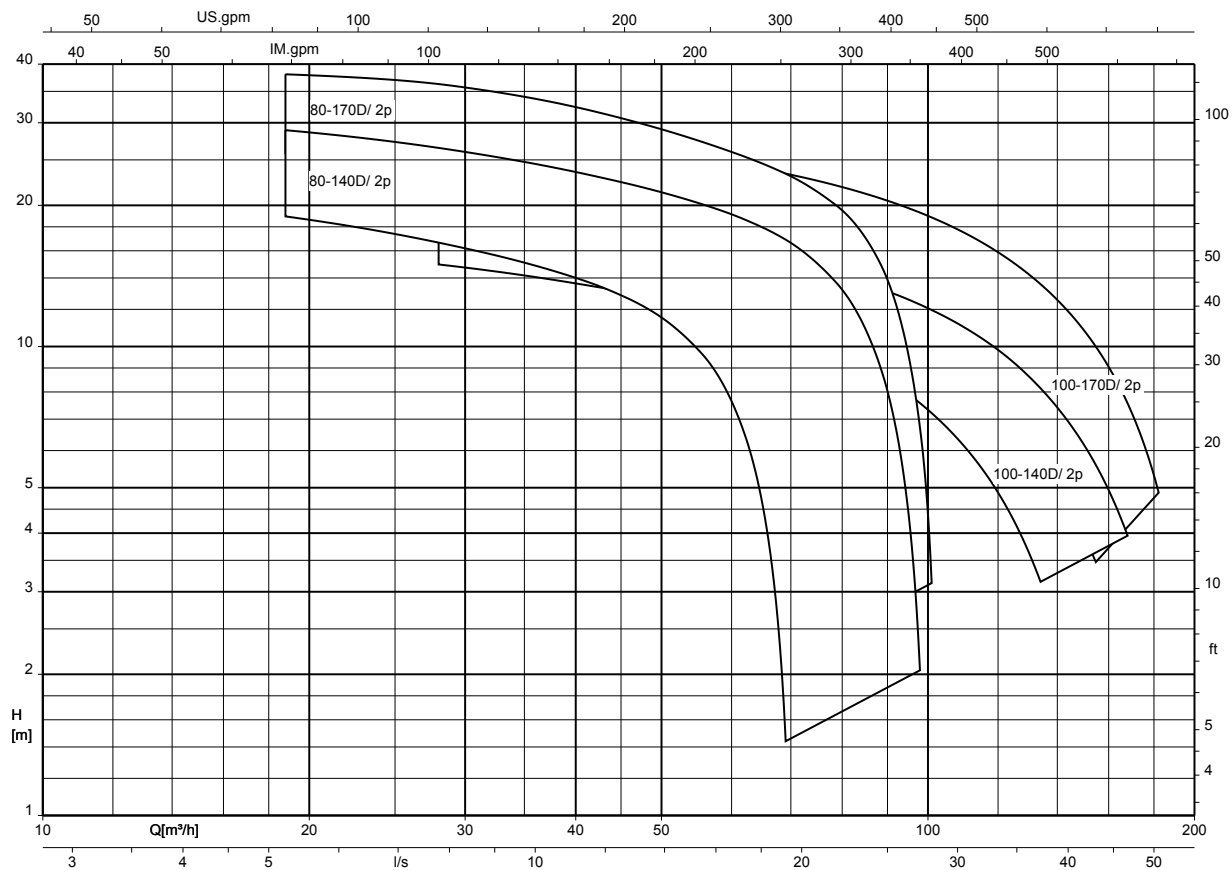
Selection charts

n = 2900 rpm

Amarex, F-max, 2-pole

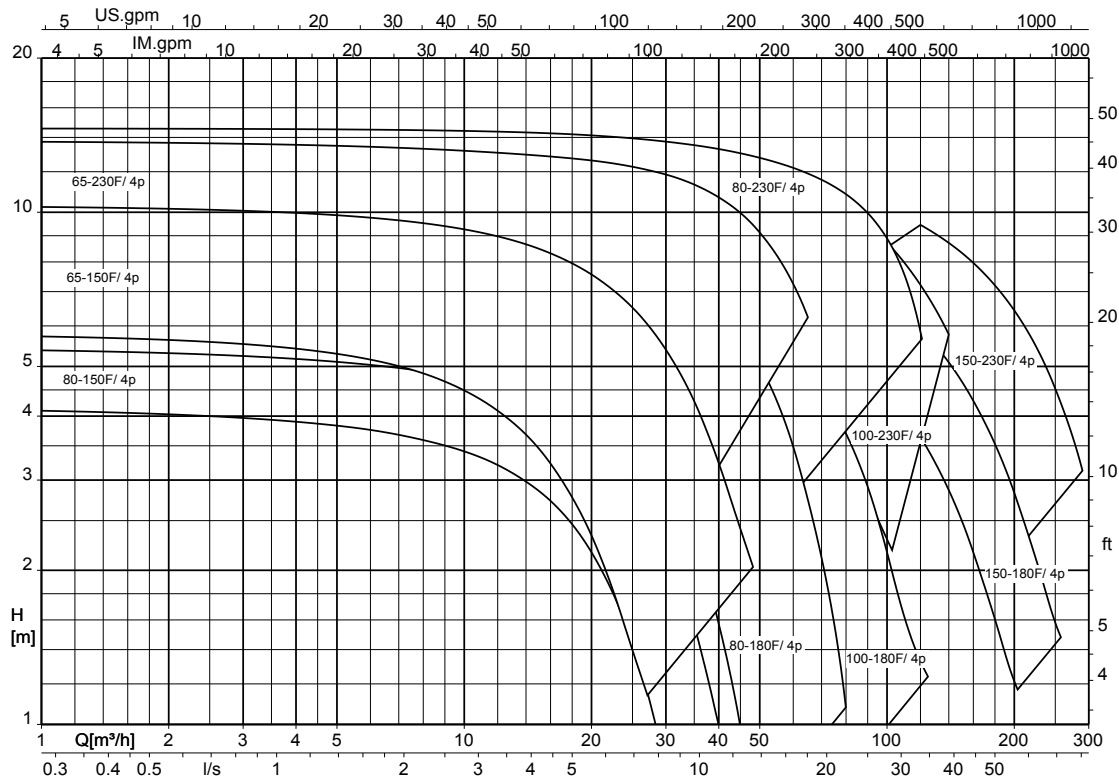


Amarex, D-max, 2-pole

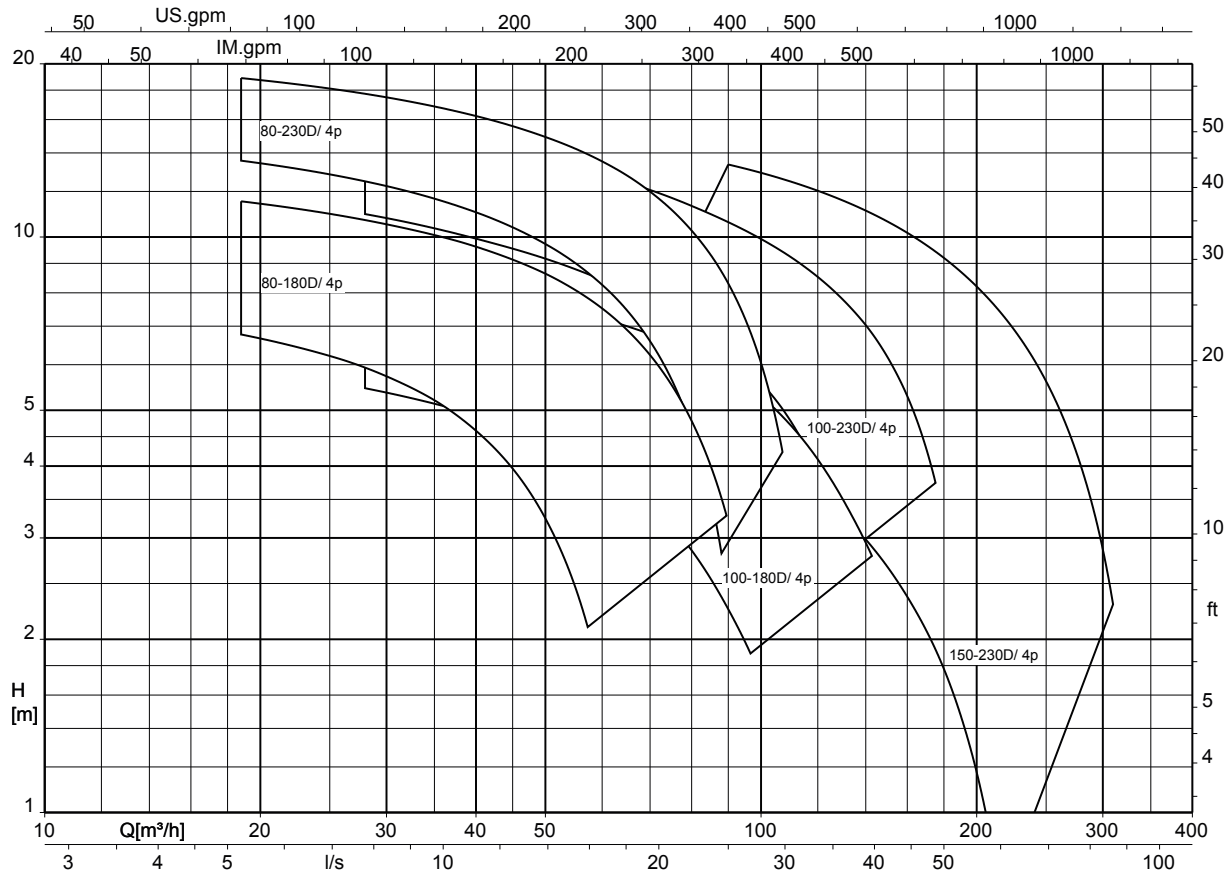


n = 1450 rpm

Amarex, F-max, 4-pole



Amarex, D-max, 4-pole



Dimensions and connections

Dimensions

Amarex DN 50/65/80/100/150, transportable version

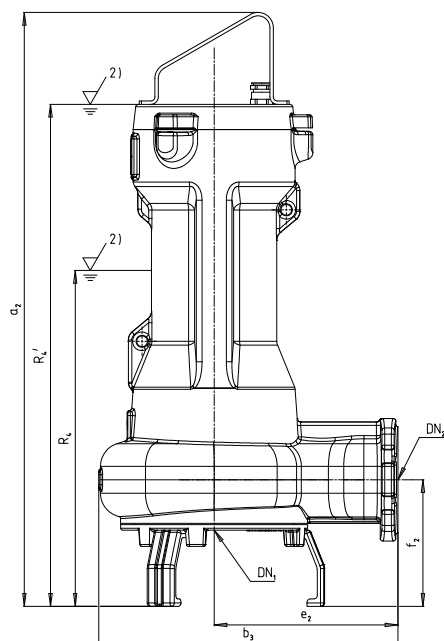


Fig. 1: Amarex, transportable version (UG1768881)

2) Minimum submergence for continuous operation

Pump dimensions [mm]

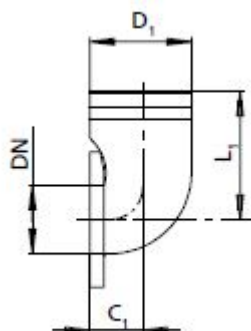
Amarex	Motor	Efficiency class	Pump											
			DN ₁	DN ₂	a ₂		b ₃	e ₂	f ₂	R ₄		R ₄ '		
					US	YS				US	YS	US	YS	
F050-140 08	014 2	C	50	50	715	724	279	170	163	345	354	-	-	
F050-220 09	023 2	F	50	50	715	724	279	170	163	-	-	600	609	
F050-220 09	018 2	C	50	50	753	762	311	180	163	378	387	-	-	
F050-220 09	024 2	F	50	50	753	762	311	180	163	-	-	638	647	
F050-220 09	029 2	C	50	50	753	762	311	180	163	423	432	-	-	
F050-220 10	035 2	C	50	50	826	835	311	180	163	381	390	-	-	
F050-220 09	040 2	F	50	50	753	762	311	180	163	-	-	638	647	
F050-220 10	045 2	C	50	50	826	835	311	180	163	381	390	-	-	
F050-220 10	051 2	F	50	50	826	835	311	180	163	-	-	691	700	
F050-220 10	073 2	F	50	50	826	835	311	180	163	-	-	691	700	
F065-150/170/230 09	017 4	F	65	65	784	793	351	210	184	-	-	669	678	
F065-150/170/230 10	021 4	C	65	65	856	865	351	210	184	391	400	-	-	
F065-150/170/230 10	035 4	F	65	65	856	865	351	210	184	-	-	721	730	
F065-150/170/230 09	018 2	C	65	65	784	793	351	210	184	369	378	-	-	
F065-150/170/230 09	024 2	F	65	65	784	793	351	210	184	-	-	669	678	
F065-150/170/230 09	029 2	C	65	65	784	793	351	210	184	454	463	-	-	
F065-150/170/230 10	035 2	C	65	65	856	865	351	210	184	391	400	-	-	
F065-150/170/230 09	040 2	F	65	65	784	793	351	210	184	-	-	669	678	
F065-150/170/230 10	045 2	C	65	65	856	865	351	210	184	411	420	-	-	
F065-150/170/230 10	051 2	F	65	65	856	865	351	210	184	-	-	721	730	
F065-150/170/230 10	073 2	F	65	65	856	865	351	210	184	-	-	721	730	
F065-150/170/230 09	012 2	C	65	65	784	793	351	210	184	369	378	-	-	
F065-150/170/230 09	017 4	F	65	65	784	793	351	210	184	-	-	669	678	
F065-150/170/230 10	021 4	C	65	65	856	865	351	210	184	358	367	-	-	
F065-150/170/230 09	023 4	F	65	65	784	793	351	210	184	-	-	669	678	
F065-150/170/230 10	035 4	F	65	65	856	865	351	210	184	-	-	721	730	

Amarex	Motor	Efficiency class	Pump										
			DN ₁	DN ₂	a ₂		b ₃	e ₂	f ₂	R ₄		R ₄ '	
					US	YS				US	YS	US	YS
F065-150/170/230 10	042 4	F	65	65	856	865	351	210	184	-	-	721	730
D080-140/180 09	018 2	C	80	80	758	767	373	230	176	343	352	-	-
D080-140/180 09	024 2	F	80	80	758	767	373	230	176	-	-	643	652
D080-140/180 09	029 2	C	80	80	758	767	373	230	176	428	437	-	-
D080-140/180 10	035 2	C	80	80	786	795	373	230	176	321	330	-	-
D080-140/180 09	040 2	F	80	80	758	767	373	230	176	-	-	643	652
D080-140/180 10	051 2	F	80	80	786	795	373	230	176	-	-	651	660
F080-150/180/220/230 09	017 4	F	80	80	798	807	385	230	194	-	-	683	692
F080-150/180/220/230 10	021 4	C	80	80	871	880	384	230	182	406	415	-	-
F080-150/180/220/230 10	035 4	F	80	80	871	880	384	230	182	-	-	736	745
D080-170/230 10	035 2	C	80	80	837	846	390	230	182	372	381	-	-
D080-170/230 10	045 2	C	80	80	837	846	390	230	182	392	401	-	-
D080-170/230 10	051 2	F	80	80	837	846	390	230	182	-	-	702	711
D080-170/230 11	060 2	C	80	80	857	866	390	230	182	432	441	-	-
D080-170/230 10	073 2	F	80	80	837	846	390	230	182	-	-	702	711
D080-170/230 11	084 2	F	80	80	857	866	390	230	182	-	-	722	731
D080-140/180 09	017 2	F	90	80	758	767	373	230	176	-	-	643	652
F080-150/180/220/230 09	017 4	F	80	80	798	807	385	230	194	-	-	683	692
D080-140/180 10	021 4	C	90	80	786	795	373	230	176	321	330	-	-
F080-150/180/220/230 10	021 4	C	80	80	871	880	384	230	182	406	415	-	-
D080-140/180 10	035 4	F	90	80	786	795	373	230	176	-	-	651	660
F080-150/180/220/230 10	035 4	F	80	80	871	880	384	230	182	-	-	736	745
F080-150/180/220/230 09	018 2	C	80	80	798	807	385	230	194	383	392	-	-
F080-150/180/220/230 09	024 2	F	80	80	798	807	385	230	194	-	-	683	692
F080-150/180/220/230 09	029 2	C	80	80	798	807	385	230	194	468	477	-	-
F080-150/180/220/230 10	035 2	C	80	80	871	880	384	230	182	406	415	-	-
F080-150/180/220/230 09	040 2	F	80	80	798	807	385	230	194	-	-	683	692
F080-150/180/220/230 10	045 2	C	80	80	871	880	384	230	182	426	435	-	-
F080-150/180/220/230 10	051 2	F	80	80	871	880	384	230	182	-	-	736	745
F080-150/180/220/230 11	060 2	C	80	80	891	900	384	230	194	466	475	-	-
F080-150/180/220/230 10	073 2	F	80	80	871	880	384	230	182	-	-	736	745
F080-150/180/220/230 11	084 2	F	80	80	891	900	384	230	194	-	-	756	765
F080-150/180/220/230 09	012 2	C	80	80	798	807	385	230	194	383	392	-	-
F080-150/180/220/230 09	017 4	F	80	80	798	807	385	230	194	-	-	683	692
D080-170/230 10	021 4	C	90	80	837	846	390	230	182	372	381	-	-
F080-150/180/220/230 10	021 4	C	80	80	871	880	384	230	182	406	415	-	-
F080-150/180/220/230 09	023 4	F	80	80	798	807	385	230	194	-	-	683	692
D080-170/230 10	035 4	F	90	80	837	846	390	230	182	-	-	702	711
F080-150/180/220/230 10	035 4	F	80	80	871	880	384	230	182	-	-	736	745
D080-170/230 11	036 4	C	90	80	857	866	390	230	182	372	381	-	-
F080-150/180/220/230 11	036 4	C	80	80	891	900	384	230	194	406	415	-	-
D080-170/230 10	042 4	F	90	80	837	846	390	230	182	-	-	702	711
F080-150/180/220/230 10	042 4	F	80	80	871	880	384	230	182	-	-	736	745
F080-150/180/220/230 11	045 4	C	80	80	891	900	384	230	194	486	495	-	-
D080-170/230 11	065 4	F	90	80	857	866	390	230	182	-	-	722	731
F080-150/180/220/230 11	065 4	F	80	80	891	900	384	230	194	-	-	756	765
F080-150/180/220/230 11	077 4	F	80	80	891	900	384	230	194	-	-	756	765
D100-140/180 09	029 2	C	90	100	784	793	431	270	193	454	463	-	-
D100-140/180 10	035 2	C	90	100	812	821	431	270	193	347	356	-	-
D100-140/180 09	040 2	F	90	100	784	793	431	270	193	-	-	669	678
D100-140/180 10	051 2	F	90	100	812	821	431	270	193	-	-	677	686
D100-170/230 10	035 2	C	90	100	844	853	440	270	186	379	388	-	-
D100-170/230 10	045 2	C	90	100	844	853	440	270	186	399	408	-	-
D100-170/230 10	051 2	F	90	100	844	853	440	270	186	-	-	709	718
D100-170/230 11	060 2	C	90	100	864	873	440	270	186	439	448	-	-
D100-170/230 10	073 2	F	90	100	844	853	440	270	186	-	-	709	718
D100-170/230 11	084 2	F	90	100	864	873	440	270	186	-	-	729	738

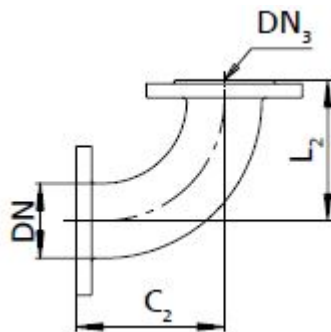
Amarex	Motor	Efficiency class	Pump											
			DN ₁	DN ₂	a ₂		b ₃	e ₂	f ₂	R ₄		R ₄ '		
					US	YS				US	YS	US	YS	
D100-140/180 09	017 4	F	110	100	784	793	431	270	193	-	-	669	678	
F100-180/220/230 09	017 4	F	100	100	821	830	433	270	207	-	-	706	715	
D100-140/180 10	021 4	C	110	100	812	821	431	270	193	347	356	-	-	
F100-180/220/230 10	021 4	C	100	100	894	903	433	270	207	429	438	-	-	
D100-140/180 10	035 4	F	110	100	812	821	431	270	193	-	-	677	686	
F100-180/220/230 10	035 4	F	100	100	894	903	433	270	207	-	-	759	768	
F100-180/220/230 10	035 2	C	100	100	894	903	433	270	207	429	438	-	-	
F100-180/220/230 10	045 2	C	100	100	894	903	433	270	207	449	458	-	-	
F100-180/220/230 10	051 2	F	100	100	894	903	433	270	207	-	-	759	768	
F100-180/220/230 11	060 2	C	100	100	914	923	433	270	207	489	498	-	-	
F100-180/220/230 10	073 2	F	100	100	894	903	433	270	207	-	-	759	768	
F100-180/220/230 11	084 2	F	100	100	914	923	433	270	207	-	-	779	788	
F100-180/220/230 09	012 2	C	100	100	821	830	433	270	207	406	415	-	-	
F100-180/220/230 09	017 4	F	100	100	821	830	433	270	207	-	-	706	715	
D100-170/230 10	021 4	C	110	100	844	853	440	270	186	379	388	-	-	
F100-180/220/230 10	021 4	C	100	100	894	903	433	270	207	429	438	-	-	
F100-180/220/230 09	023 4	F	100	100	821	830	433	270	207	-	-	706	715	
D100-170/230 10	035 4	F	110	100	844	853	440	270	186	-	-	709	718	
F100-180/220/230 10	035 4	F	100	100	894	903	433	270	207	-	-	759	768	
D100-170/230 11	036 4	C	110	100	864	873	440	270	186	379	388	-	-	
F100-180/220/230 11	036 4	C	100	100	914	923	433	270	207	429	438	-	-	
D100-170/230 10	042 4	F	110	100	844	853	440	270	186	-	-	709	718	
F100-180/220/230 10	042 4	F	100	100	894	903	433	270	207	-	-	759	768	
F100-180/220/230 11	045 4	C	100	100	914	923	433	270	207	509	518	-	-	
D100-170/230 11	065 4	F	110	100	864	873	440	270	186	-	-	729	738	
F100-180/220/230 11	065 4	F	100	100	914	923	433	270	207	-	-	779	788	
F100-180/220/230 11	077 4	F	100	100	914	923	433	270	207	-	-	779	788	
F150-180/230 11	036 4	C	150	150	926	935	512	320	217	441	450	-	-	
F150-180/230 11	065 4	F	150	150	926	935	512	320	217	-	-	791	800	
D150-230 11	036 4	C	150	150	926	935	512	320	217	441	450	-	-	
F150-180/230 11	036 4	C	150	150	926	935	512	320	217	441	450	-	-	
D150-230 11	045 4	C	150	150	926	935	512	320	217	521	530	-	-	
F150-180/230 11	045 4	C	150	150	926	935	512	320	217	521	530	-	-	
D150-230 11	065 4	F	150	150	926	935	512	320	217	-	-	791	800	
F150-180/230 11	065 4	F	150	150	926	935	512	320	217	-	-	-	800	
D150-230 11	077 4	F	150	150	926	935	512	320	217	-	-	-	800	
F150-180/230 11	077 4	F	150	150	926	935	512	320	217	-	-	-	800	

Connection elbow

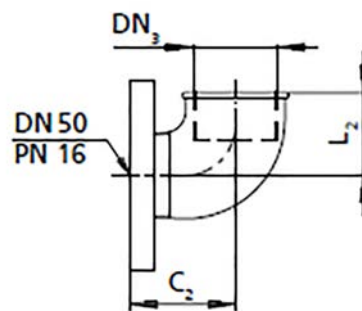
Connection elbow with flange/hose connection (P13)



Connection elbow with flanges (P14)



Connection elbow with internal thread and external thread (P14) and threaded flange (P27)

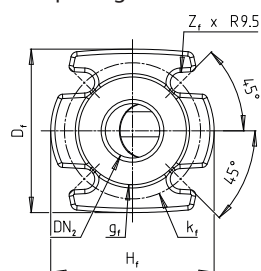


Elbow dimensions [mm]

DN	Connection elbow with flange/hose connection (P13)			Connection elbow with flanges (P14)			Connection elbow with internal thread and external thread (P14) and threaded flange (P27)		
	D ₁	C ₁	L ₁	DN ₃	C ₂	L ₂	DN ₃	C ₂	L ₂
50	-	-	-	-	-	-	G 2"	78	58
65	75	40	135	65	135	135	-	-	-
80	75	115	175	80	135	135	-	-	-
100	110	45	195	100	120	135	-	-	-

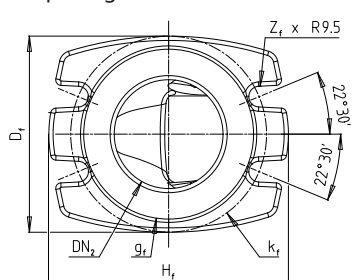
Pump flange DN₂

Pump flange DN 50 / 65



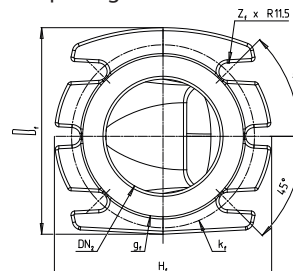
(UG1748881)

Pump flange DN 80 / 100



(UG1748881)

Pump flange DN 150



(UG1748881)

Flange dimensions [mm]

Flange variant	DN ₂	g _f	k _f	D _f	H _f	Z _f
DN 50/50 ISO	50	102	160	125	150	4
DN 50/50 ASME	50	102	160	120,5	150	4
DN 65/65	65	122	140	140	170	4
DN 65/80 ISO	80	133	160	200	170	8
DN 65/80 ASME	80	127	152	191	170	4
DN 80/80 ISO	80	132	160	200	200	8
DN 80/80 ASME	80	127	152	190	200	4
DN 80/100	100	156	180	220	200	8
DN 100/100	100	156	180	200	220	8
DN 150/150	100	212	240	285	285	8

Amarex DN 50, stationary installation, guide wire arrangement, single guide rail arrangement, guide hoop arrangement

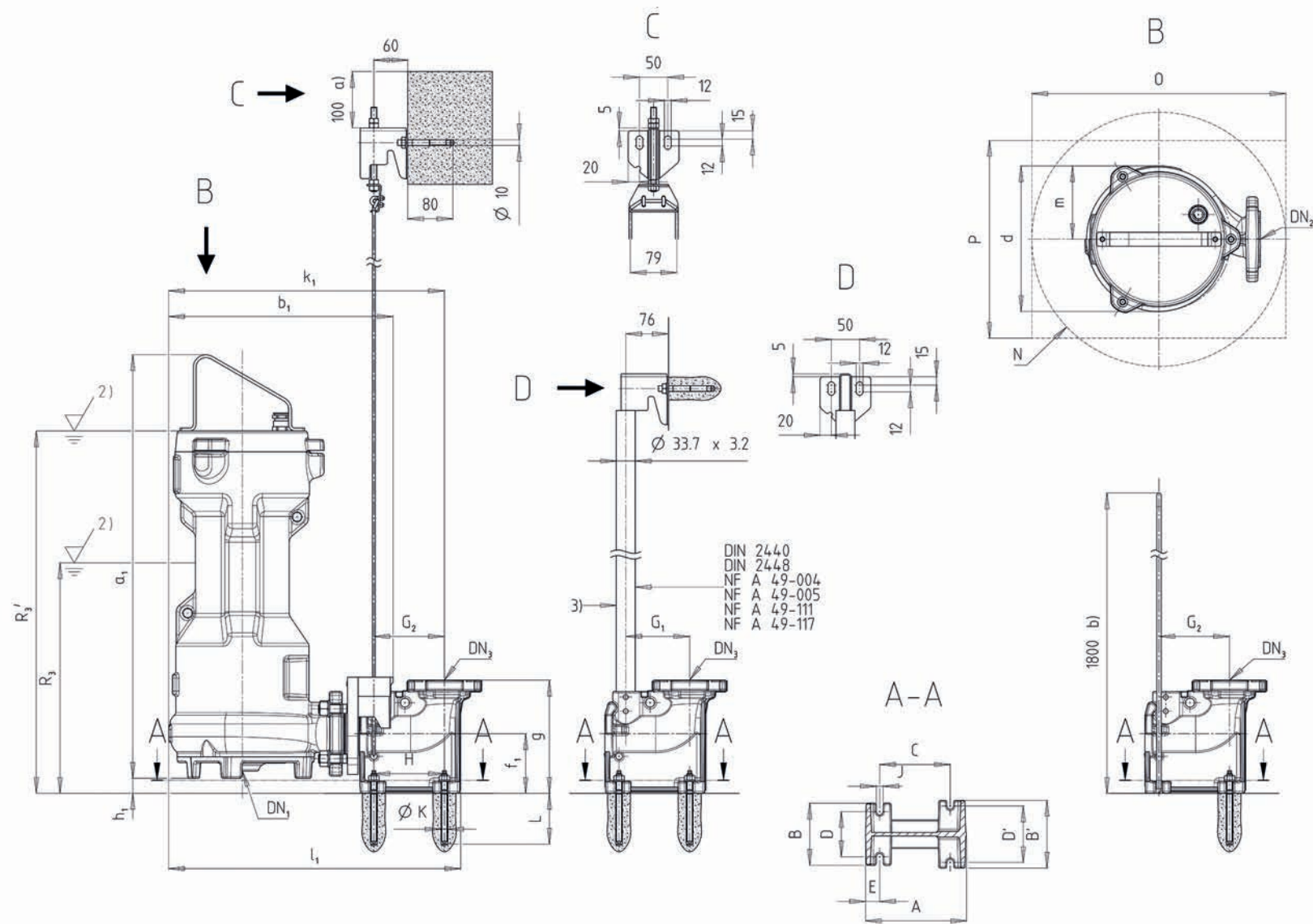
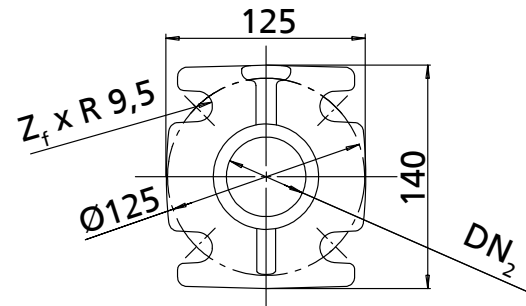


Fig. 2: Amarex 50, stationary installation, guide wire arrangement, single guide rail arrangement, guide hoop arrangement (UG1752288)

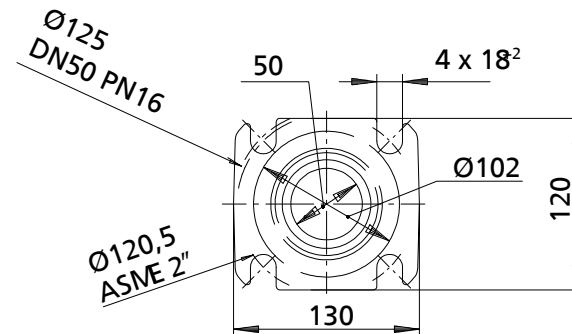
1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

Dimensions of pump set and foundation [mm]

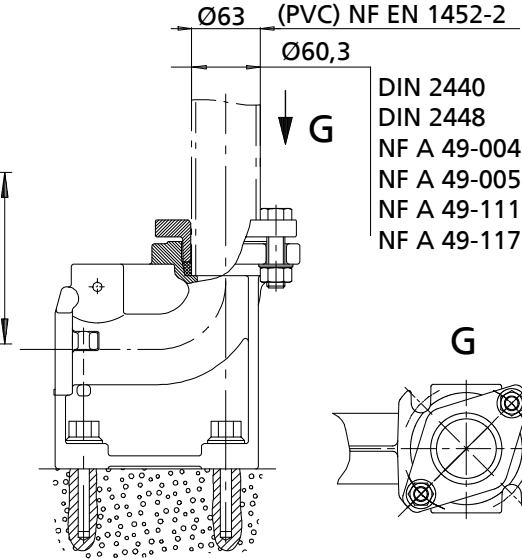
Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	min.	min.															min.		
F050-140	014 2	C	42	50	631	640	368	216	150	106	201	99	27	150	457	125	486	109	4	288	297	-	-	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-140	023 2	F	42	50	631	640	368	216	150	106	201	99	27	150	457	125	486	109	4	-	-	543	552	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	017 2	C	42	50	669	678	398	258	150	106	201	99	27	150	489	125	518	130	4	321	330	-	-	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	024 2	F	42	50	669	678	398	258	150	106	201	99	27	150	489	125	518	130	4	-	-	581	590	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	029 2	C	42	50	669	678	398	258	150	106	201	99	27	150	489	125	518	130	4	366	375	-	-	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	035 2	C	42	50	742	751	398	258	150	106	201	99	27	150	489	125	518	130	4	324	333	-	-	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	040 2	F	42	50	669	678	398	258	150	106	201	99	27	150	489	125	518	130	4	-	-	581	590	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	045 2	C	42	50	742	751	398	258	150	106	201	99	27	150	489	125	518	130	4	324	333	-	-	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	051 2	F	42	50	742	751	398	258	150	106	201	99	27	150	489	125	518	130	4	-	-	634	643	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350
F050-220	073 2	F	42	50	742	751	398	258	150	106	201	99	27	150	489	125	518	130	4	-	-	561	570	50	179	110	120	125	80	100	25	113	125	125	12	10	90	465	465	350

Pump flange DN₂Flanged bend DN₃

ASME 2"
ASME B16.1 class 125



Clamped connection



Amarex DN 50, stationary installation, twin guide rail arrangement

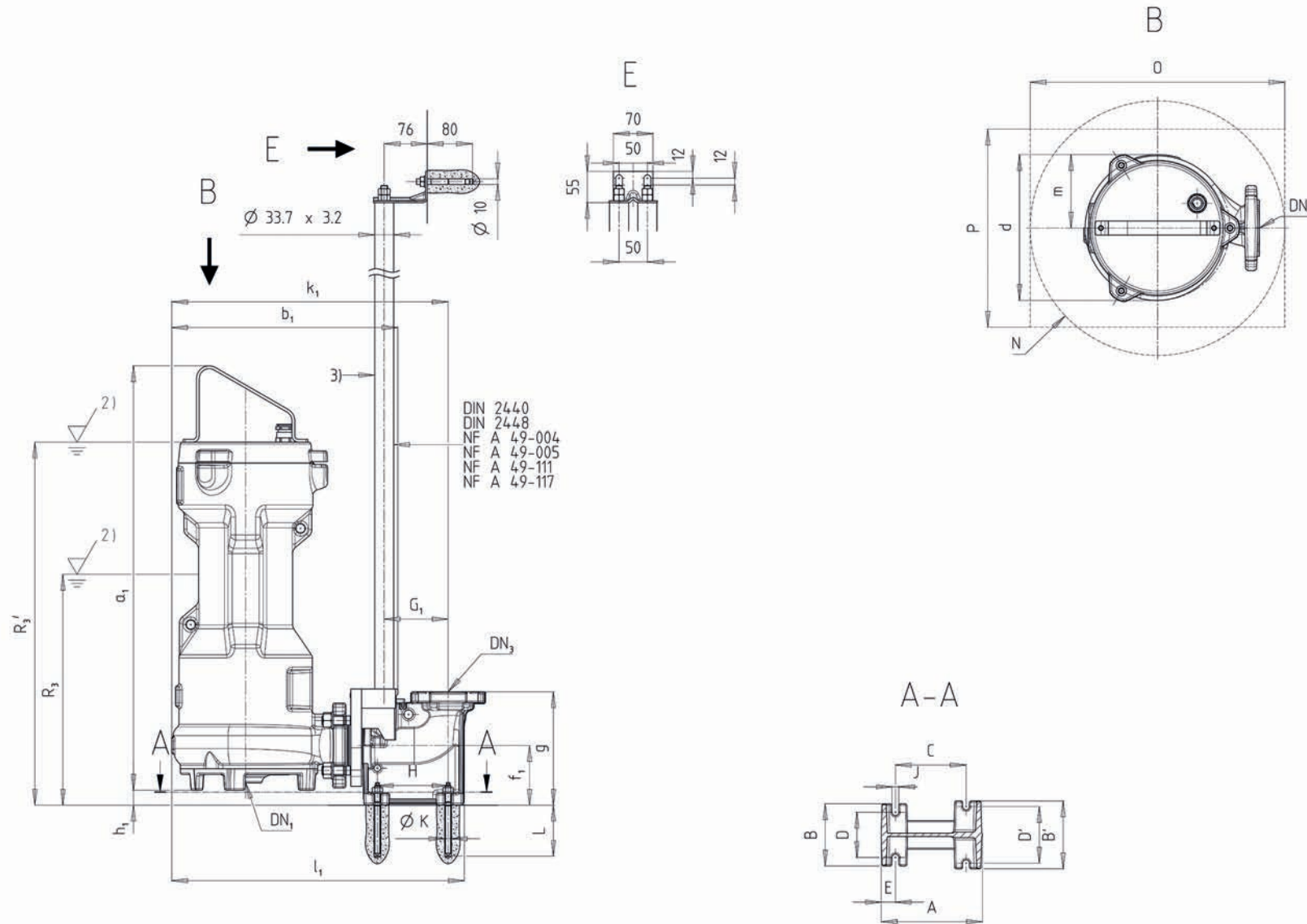
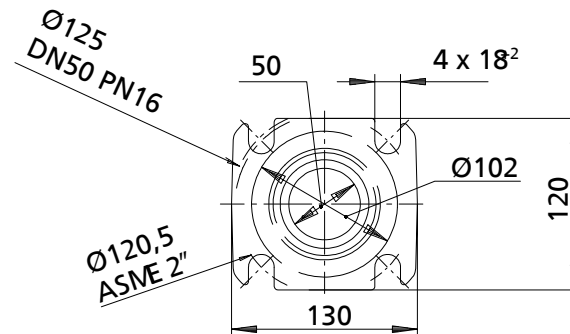


Fig. 3: Amarex 50, stationary installation, twin guide rail arrangement (UG1752288)

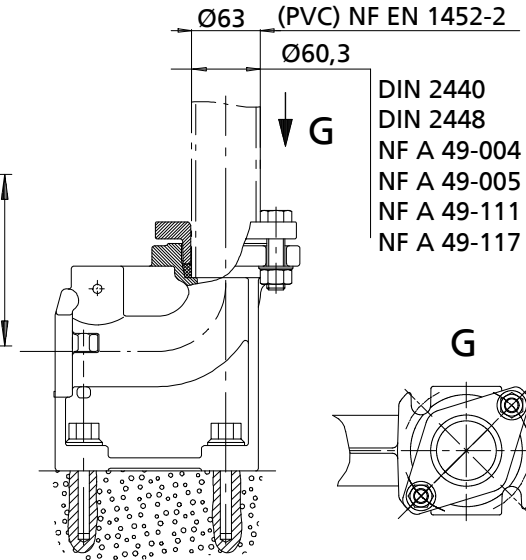
1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS															min.	min.	min.
F050-140	014 2	C	42	50	631	640	368	216	150	106	201	99	27	150	457	125	486	109	4	288	297	-	-	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-140	023 2	F	42	50	631	640	368	216	150	106	201	99	27	150	457	125	486	109	4	-	-	543	552	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	017 2	C	42	50	669	678	400	258	150	106	201	99	27	150	489	125	518	130	4	321	330	-	-	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	024 2	F	42	50	669	678	400	258	150	106	201	99	27	150	489	125	518	130	4	-	-	581	590	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	029 2	C	42	50	669	678	400	258	150	106	201	99	27	150	489	125	518	130	4	366	375	-	-	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	035 2	C	42	50	742	751	400	258	150	106	201	99	27	150	489	125	518	130	4	324	333	-	-	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	040 2	F	42	50	669	678	400	258	150	106	201	99	27	150	489	125	518	130	4	-	-	581	590	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	045 2	C	42	50	742	751	400	258	150	106	201	99	27	150	489	125	518	130	4	324	333	-	-	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	051 2	F	42	50	742	751	400	258	150	106	201	99	27	150	489	125	518	130	4	-	-	634	643	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350
F050-220	073 2	F	42	50	742	751	400	258	150	106	201	99	27	150	489	125	518	130	4	-	-	561	570	50	179	110	120	125	80	100	25	113	-	125	12	10	90	465	465	350

Flanged bend DN₃ASME 2"
ASME B16.1 class 125

Clamped connection



Amarex DN 65, stationary installation, guide wire arrangement, single guide rail arrangement, guide hoop arrangement

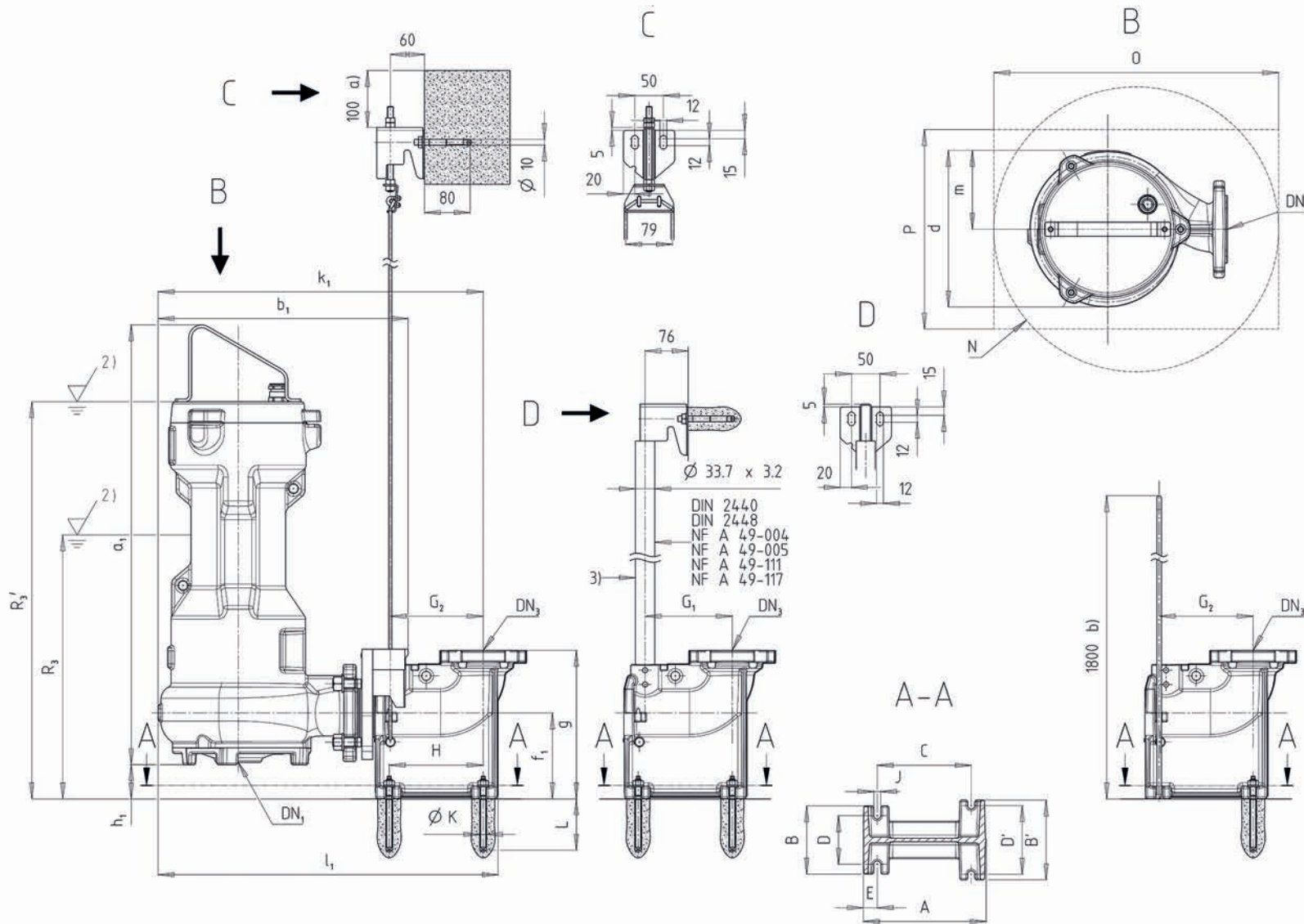


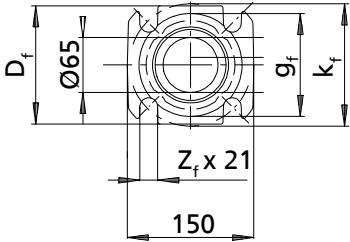
Fig. 4: Amarex 65, stationary installation, guide wire arrangement, single guide rail arrangement, guide hoop arrangement (UG1751861)

1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

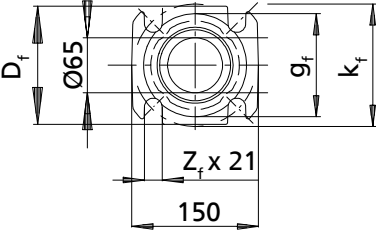
Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS																			
																						min.	min.																	
F065-150	017 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-150	021 4	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-150	035 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	018 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	336	345	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	024 2	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	029 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	421	430	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	035 2	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	040 2	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	045 2	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	378	387	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	051 2	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	073 2	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	012 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	336	345	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	017 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	021 4	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	023 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	035 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	042 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400

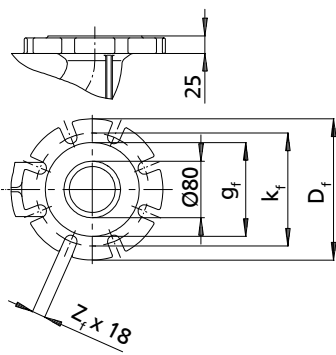
Flanged bend DN₃
DN 65/65



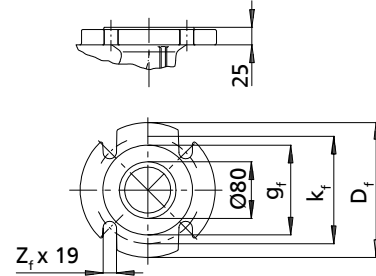
DN 65/65
ASME B 16.1 Class 125
2 1/2"



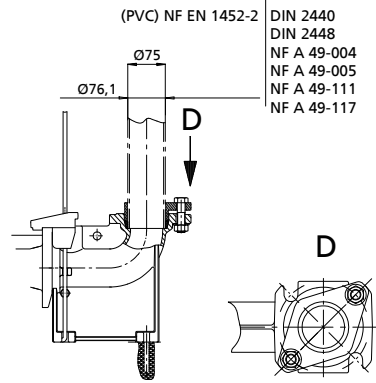
Extension elbow DN₃
DN 65/80



DN 65/80
ASME B 16.1 Class 125
3"



Clamped connection



Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	gf	kf	Df	Zf
ISO 7005 PN16	65	122	145	140	4
DIN 2501 PN16	80	133	160	200	8
ASME B16.1 class 125	65	122	140	140	4
	80	127	152,5	191	4

Amarex DN 65, stationary installation, twin guide rail arrangement

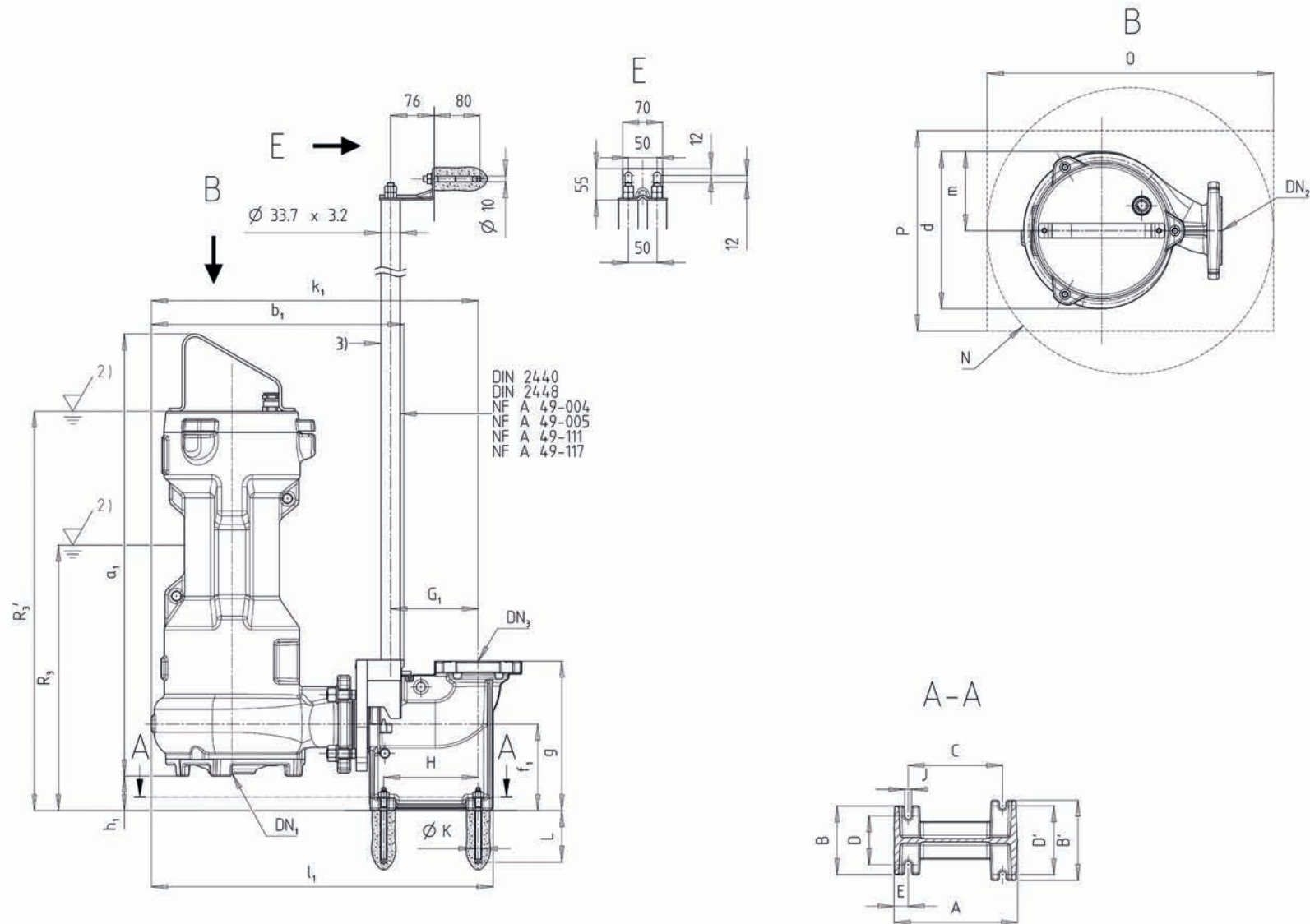


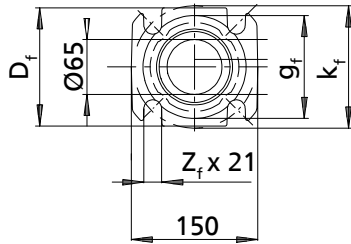
Fig. 5: Amarex 65, stationary installation, twin guide rail arrangement (UG1751861)

1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

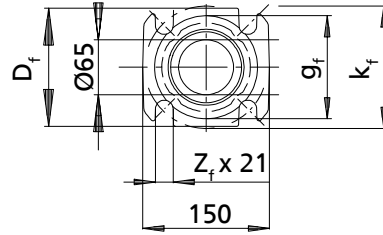
Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z ₁	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	min.	min.															min.		
F065-150	017 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-150	021 4	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-150	035 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	018 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	336	345	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	024 2	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	029 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	421	430	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	035 2	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	040 2	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	045 2	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	378	387	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	051 2	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-170	073 2	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	012 2	C	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	336	345	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	017 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	021 4	C	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	358	367	-	-	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	023 4	F	65	65	691	700	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	636	645	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	035 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400
F065-230	042 4	F	65	65	763	772	440	274	170	151	261	118	60	170	569	145	595	138	4	-	-	688	697	65	216	120	140	165	85	120	25	153	-	165	12	10	90	550	550	400

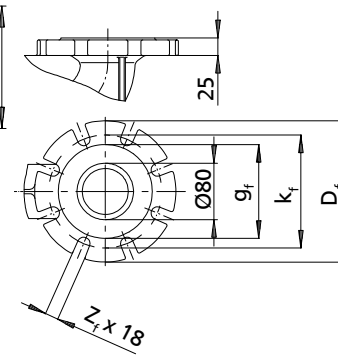
Flanged bend DN₃
DN 65/65



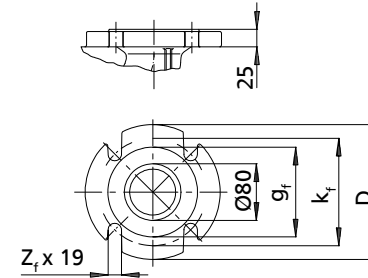
DN 65/65
ASME B 16.1 Class 125
2 1/2"



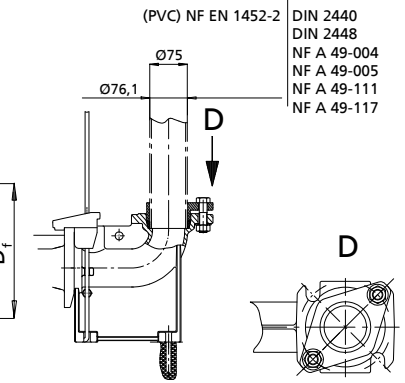
Extension elbow DN₃
DN 65/80



DN 65/80
ASME B 16.1 Class 125
3"



Clamped connection



(PVC) NF EN 1452-2
DIN 2440
DIN 2448
NF A 49-004
NF A 49-005
NF A 49-111
NF A 49-117

Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	gf	kf	Df	Zf
ISO 7005 PN16	65	122	145	140	4
DIN 2501 PN16	80	133	160	200	8
ASME B16.1 class 125	65	122	140	140	4
	80	127	152,5	191	4

Amarex DN 80, stationary installation, guide wire arrangement, twin guide rail arrangement

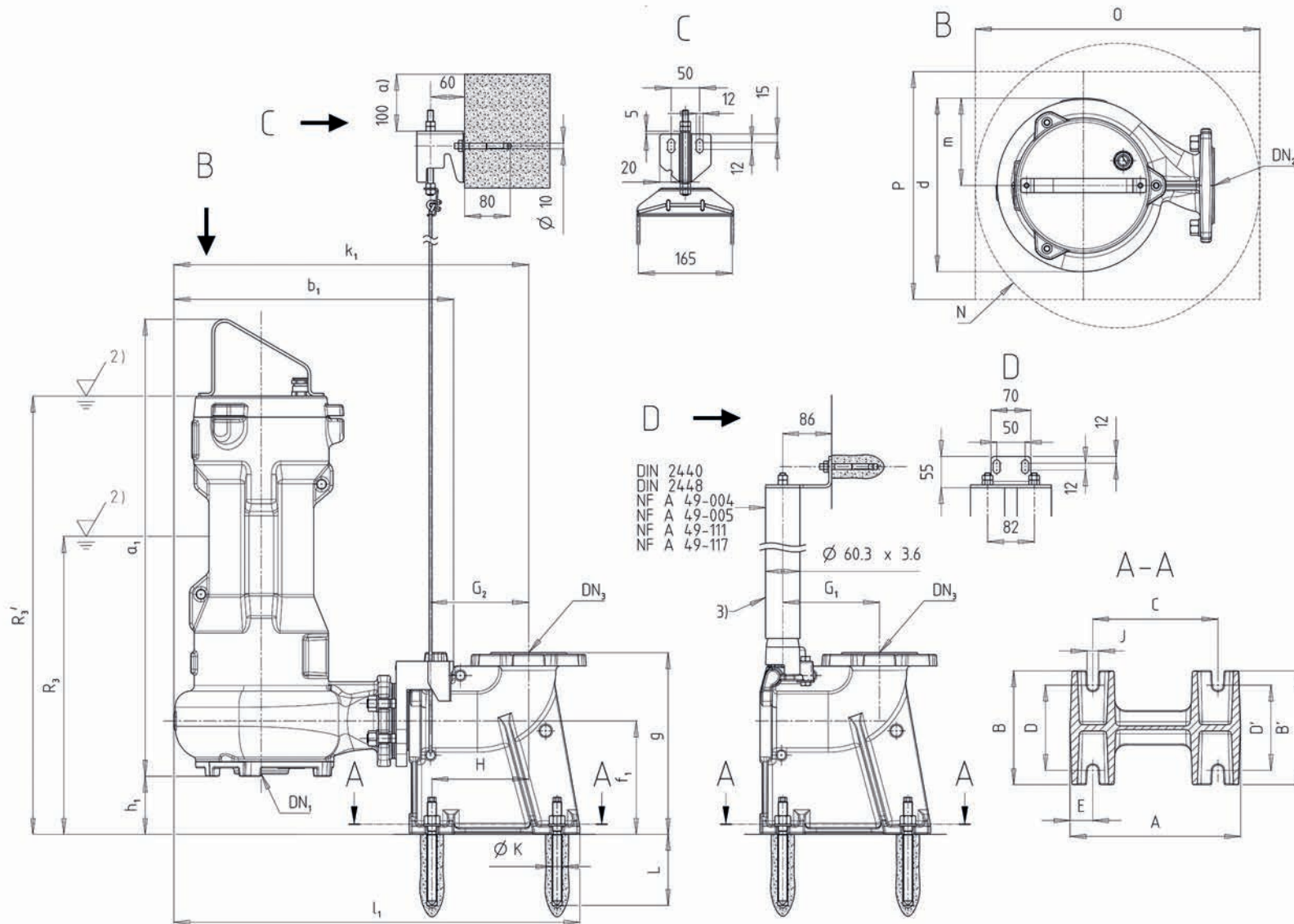


Fig. 6: Amarex 80, stationary installation, guide wire arrangement, twin guide rail arrangement (UG1751843)

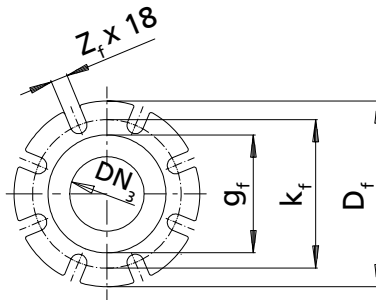
1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

Dimensions of pump set and foundation [mm]

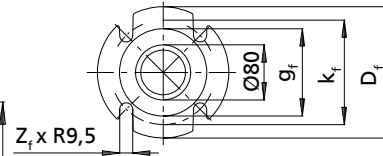
Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _r	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS																	
D080-140	018 2	C	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	367	376	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-140	024 2	F	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-140	029 2	C	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	452	461	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-140	035 2	C	80	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	345	354	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-140	040 2	F	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-140	051 2	F	80	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	675	684	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-150	017 4	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-150	021 4	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-150	035 4	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	035 2	C	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	390	399	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	045 2	C	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	410	419	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	051 2	F	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	720	729	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	060 2	C	80	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	450	459	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	073 2	F	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	720	729	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-170	084 2	F	80	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	740	749	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-180	017 2	F	90	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-180	017 4	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-180	021 4	C	90	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	345	354	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-180	021 4	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
D080-180	035 4	F	90	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	675	684	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-180	035 4	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	018 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	389	398	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	024 2	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	029 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	474	483	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	035 2	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	040 2	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	045 2	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	452	461	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	051 2	F	90	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	060 2	C	80	80	794	803	491	35	160	200	320	132	103	200	622	160	712	154	4	492	501	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	073 2	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-220	084 2	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-230	012 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	389	398	-	-	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	

Anarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	min.	min.															min.		
D080-230	065 4	F	90	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	740	749	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-230	065 4	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400
F080-230	077 4	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	170	173	170	20	18	125	580	580	400

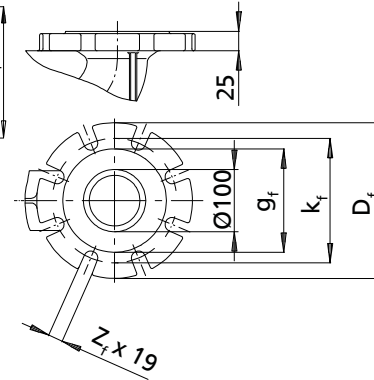
Flanged bend DN₃ DN 80/80



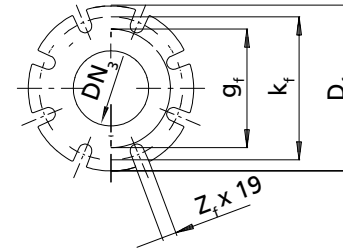
DN 80/80 ASME B 16.1 Class 125 3"



Extension elbow DN₃ DN 80/80



DN 80/100 ASME B 16.1 Class 125 4"

Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 150 lbs	80	127	152,5	191	4
	100	156	190,5	220	8

Amarex DN 80, stationary installation, single guide rail arrangement

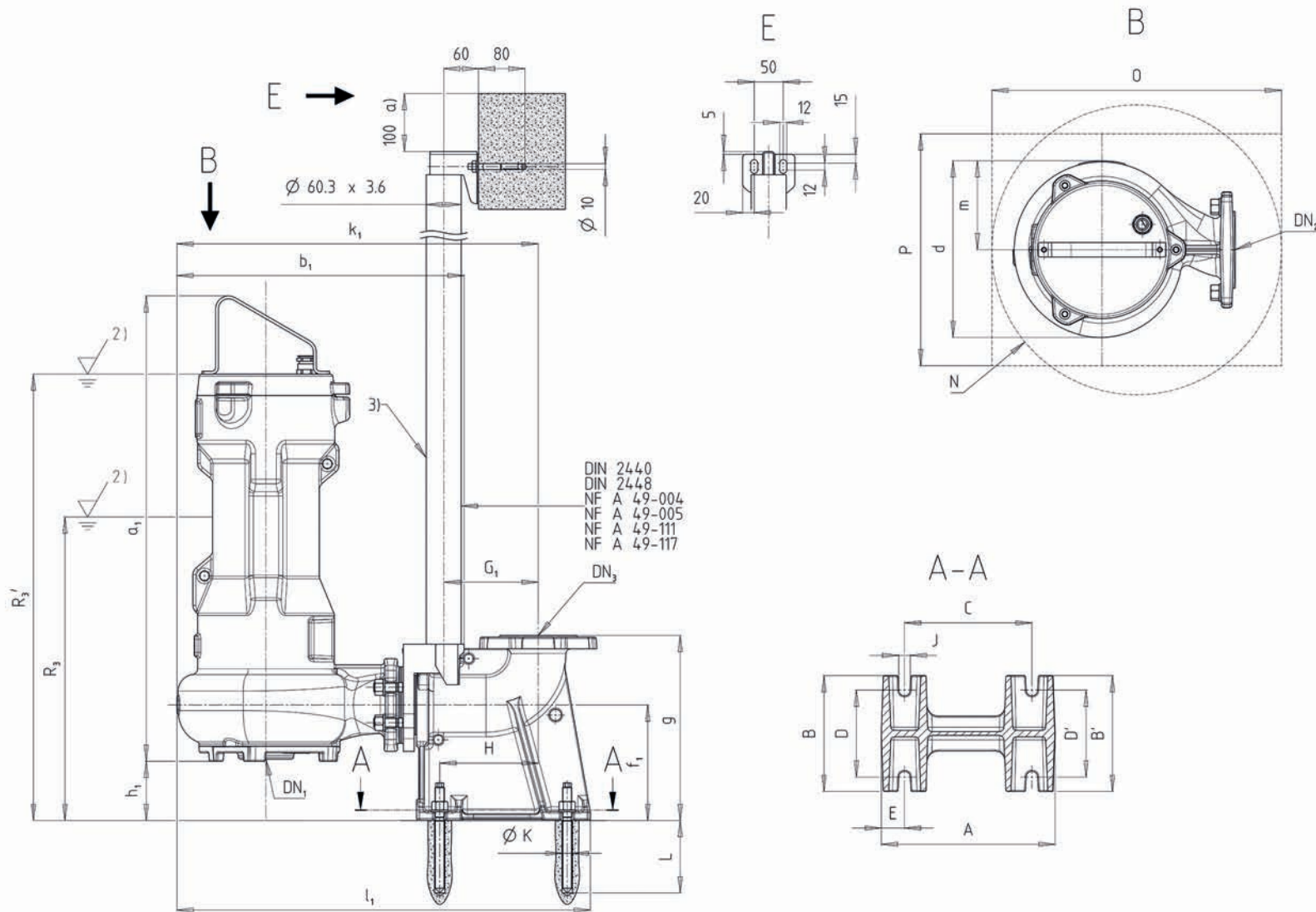


Fig. 7: Amarex 80, stationary installation, single guide rail arrangement (UG17518431)

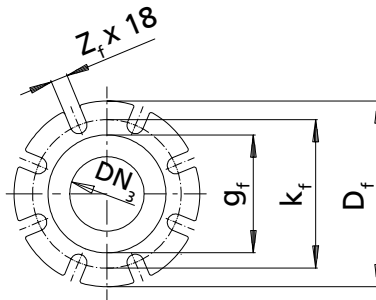
1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

Dimensions of pump set and foundation [mm]

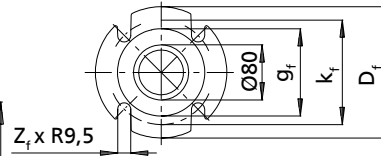
Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																					
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃				R ₃ '				DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS	US	YS	US	YS																	
D080-140	018 2	C	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	367	376	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-140	024 2	F	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-140	029 2	C	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	452	461	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-140	035 2	C	80	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	345	354	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-140	040 2	F	80	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-140	051 2	F	80	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	675	684	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-150	017 4	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-150	021 4	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-150	035 4	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	035 2	C	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	390	399	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	045 2	C	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	410	419	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	051 2	F	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	720	729	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	060 2	C	80	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	450	459	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	073 2	F	80	80	743	752	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	720	729	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-170	084 2	F	80	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	740	749	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-180	017 2	F	90	80	670	679	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	667	676	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-180	017 4	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-180	021 4	C	90	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	345	354	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-180	021 4	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
D080-180	035 4	F	90	80	698	707	480	282	160	200	320	132	112	200	611	160	701	154	4	-	-	675	684	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-180	035 4	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	018 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	389	398	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	024 2	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	029 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	474	483	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	035 2	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	432	441	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	040 2	F	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	-	-	689	698	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	045 2	C	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	452	461	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	051 2	F	90	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	060 2	C	80	80	794	803	491	35	160	200	320	132	103	200	622	160	712	154	4	492	501	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	073 2	F	80	80	774	783	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	762	771	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-220	084 2	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-230	012 2	C	80	80	701	710	492	305	160	200	320	132	103	200	623	160	713	154	4	389	398	-	-	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400				
F080-230	017 4	F	80	80	701																																							

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS															min.	min.	min.
D080-230	065 4	F	90	80	763	772	497	311	160	200	320	132	112	200	628	160	718	166	4	-	-	740	749	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400
F080-230	065 4	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400
F080-230	077 4	F	80	80	794	803	491	305	160	200	320	132	103	200	622	160	712	154	4	-	-	782	791	80	300	200	200	220	150	150	40	163	-	170	20	18	125	580	580	400

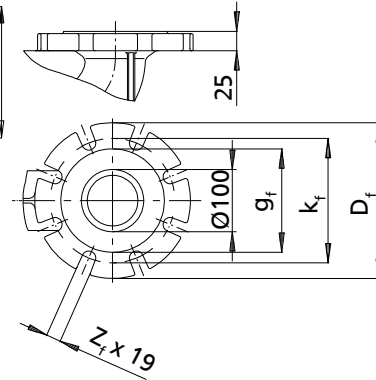
Flanged bend DN₃
DN 80/80



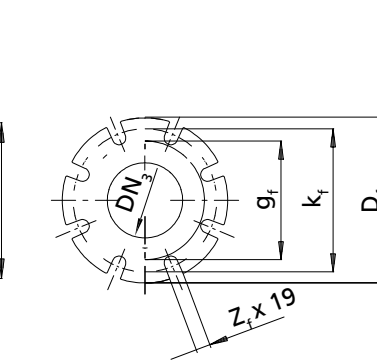
DN 80/80
ASME B 16.1 Class 125
3"



Extension elbow DN₃
DN 80/80



DN 80/100
ASME B 16.1 Class 125
4"



Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 150 lbs	80	127	152,5	191	4
	100	156	190,5	220	8

Amarex DN 100, stationary installation, guide wire arrangement, twin guide rail arrangement

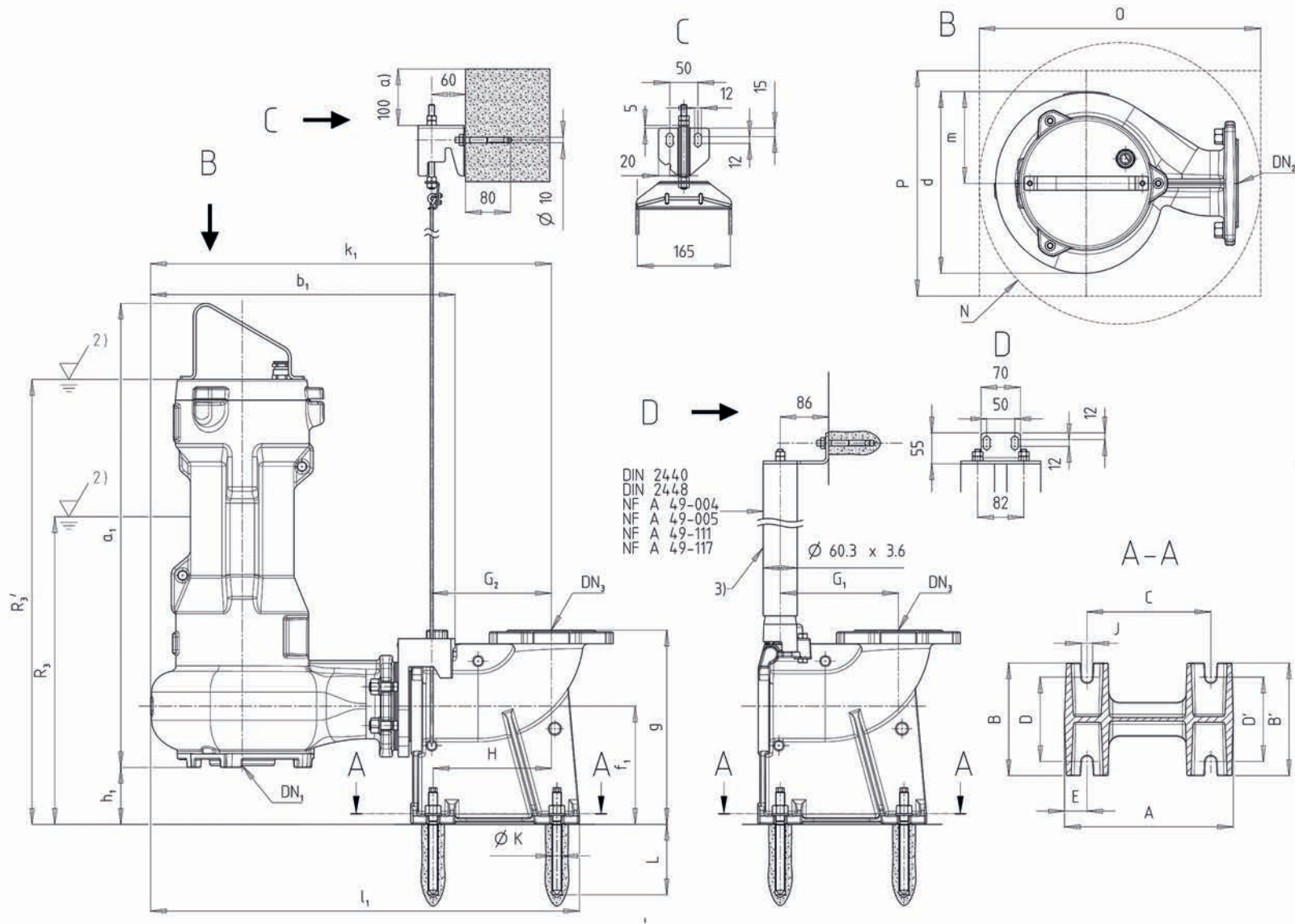


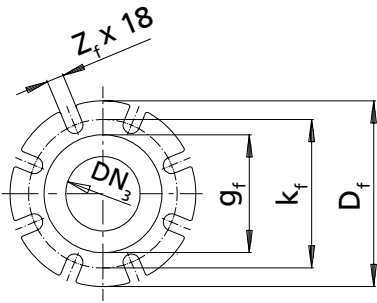
Fig. 8: Amarex 100, stationary installation, guide wire arrangement, twin guide rail arrangement (UG1751626)

1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

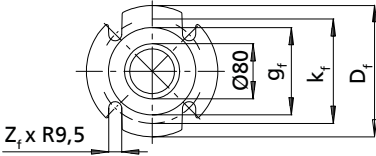
Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _r	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	min.	min.															min.		
D100-140	029 2	C	90	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	471	480	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-140	035 2	C	90	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	364	373	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-140	040 2	F	90	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	686	695	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-140	051 2	F	90	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	694	703	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	035 2	C	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	045 2	C	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	423	432	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	051 2	F	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	060 2	C	90	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	463	472	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	073 2	F	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-170	084 2	F	90	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	753	762	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-180	017 4	F	110	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	686	695	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-180	017 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-180	021 4	C	110	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	364	373	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-180	021 4	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-180	035 4	F	110	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	694	703	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-180	035 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	035 2	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	045 2	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	452	461	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	051 2	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	060 2	C	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	492	501	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	073 2	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-220	084 2	F	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	782	791	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	012 2	C	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	409	418	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	017 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-230	021 4	C	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	021 4	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	023 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-230	035 4	F	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	035 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-230	036 4	C	110	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	036 4	C	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
D100-230	042 4	F	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	210	213	210	20	18	125	650	650	420
F100-230	042 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760</																							

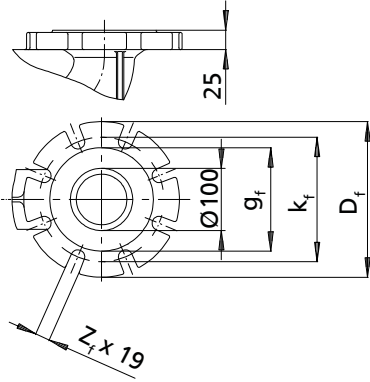
Flanged bend DN₃
DN 80/80



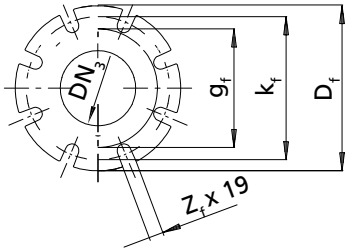
DN 80/80
ASME B 16.1 Class 125
3"



Extension elbow DN₃
DN 80/80



DN 80/100
ASME B 16.1 Class 125
4"



Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 150 lbs	80	127	152,5	191	4
	100	156	190,5	220	8

Amarex DN 100, stationary installation, single guide rail arrangement

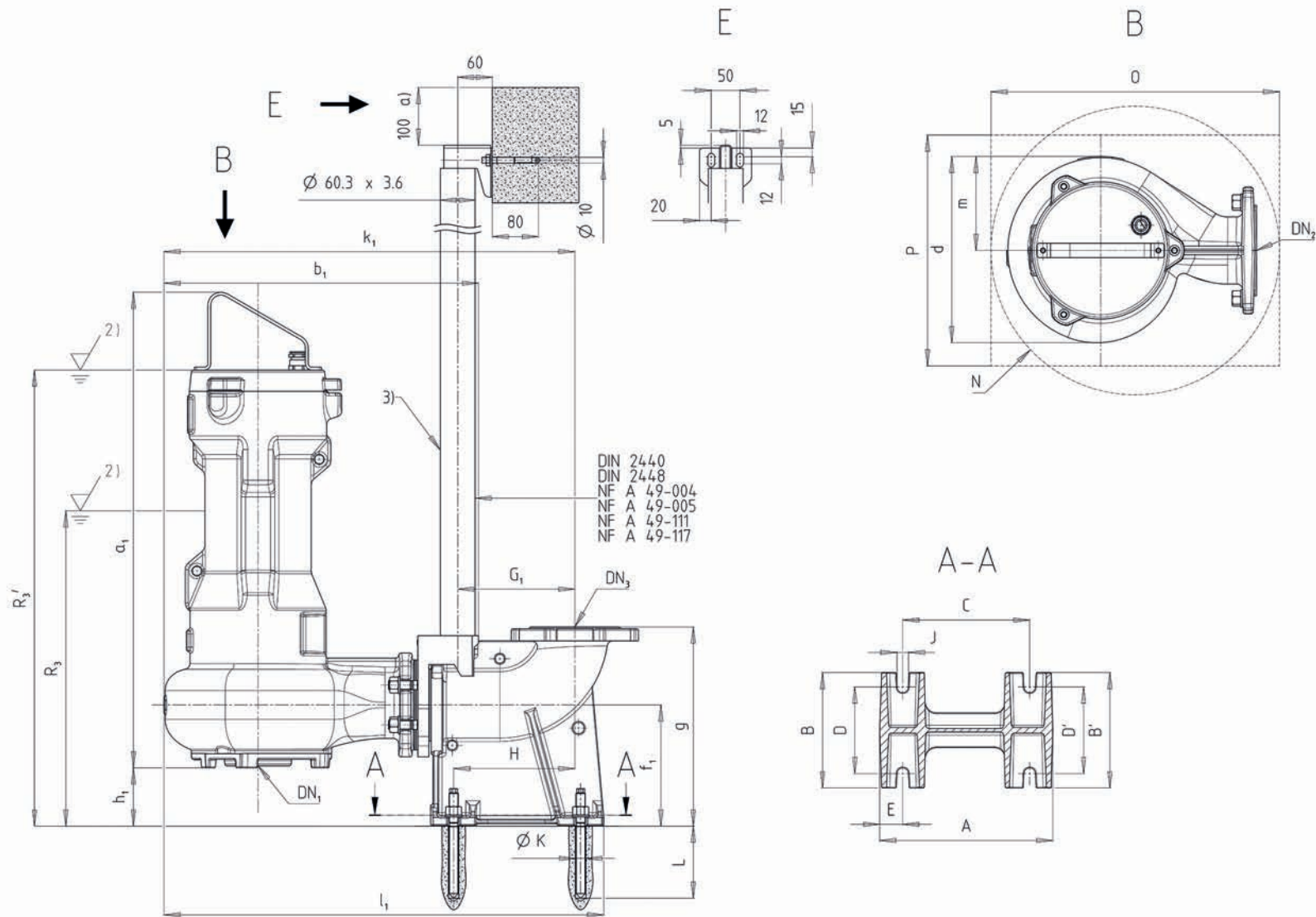


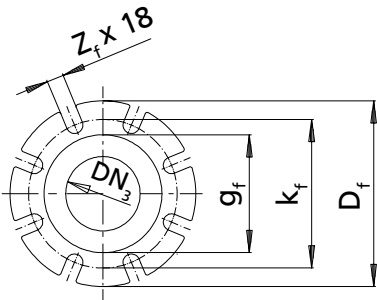
Fig. 9: Amarex 100, stationary installation, single guide rail arrangement (UG1751626)

1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

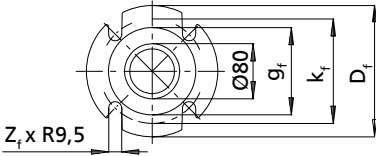
Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																				Foundation (installation in a sump)																	
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _r	h ₁	H _f	k ₁	k _r	l ₁	m	Z _r	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS																	
D100-140	029 2	C	90	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	471	480	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-140	035 2	C	90	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	364	373	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-140	040 2	F	90	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	686	695	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-140	051 2	F	90	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	694	703	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	035 2	C	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	045 2	C	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	423	432	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	051 2	F	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	060 2	C	90	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	463	472	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	073 2	F	90	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-170	084 2	F	90	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	753	762	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-180	017 4	F	110	100	689	698	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	686	695	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-180	017 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-180	021 4	C	110	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	364	373	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-180	021 4	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-180	035 4	F	110	100	717	726	538	324	180	210	345	156	112	220	708	180	758	185	4	-	-	694	703	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-180	035 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	035 2	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	045 2	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	452	461	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	051 2	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	060 2	C	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	492	501	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	073 2	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-220	084 2	F	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	782	791	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	012 2	C	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	409	418	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	017 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-230	021 4	C	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	021 4	C	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	023 4	F	100	100	724	733	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	709	718	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-230	035 4	F	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	035 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-230	036 4	C	110	100	776	785	547	336	180	210	345	156	112	220	717	180	767	189	4	403	412	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	036 4	C	100	100	817	826	540	323	180	210	345	156	100	220	710	180	760	163	4	432	441	-	-	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
D100-230	042 4	F	110	100	756	765	547	336	180	210	345	156	112	220	717	180	767	189	4	-	-	733	742	100	300	200	200	220	150	150	40	203	-	210	20	18	125	650	650	420
F100-230	042 4	F	100	100	797	806	540	323	180	210	345	156	100	220	710	180	760	163	4	-	-	762	771	100	300</															

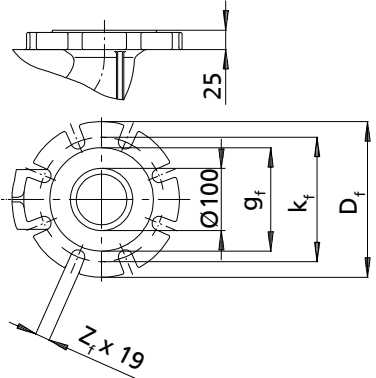
Flanged bend DN₃
DN 80/80



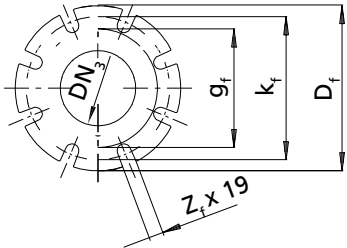
DN 80/80
ASME B 16.1 Class 125
3"



Extension elbow DN₃
DN 80/80



DN 80/100
ASME B 16.1 Class 125
4"



Dimensions of flanged bend DN₃ [mm]

Flange design	DN ₃	g _f	k _f	D _f	Z _f
ISO 7005 PN16	80	132	160	200	8
DIN 2501 PN16	100	156	180	220	8
ASME B16.1 150 lbs	80	127	152,5	191	4
	100	156	190,5	220	8

Amarex DN 150, stationary installation, guide wire arrangement, twin guide rail arrangement

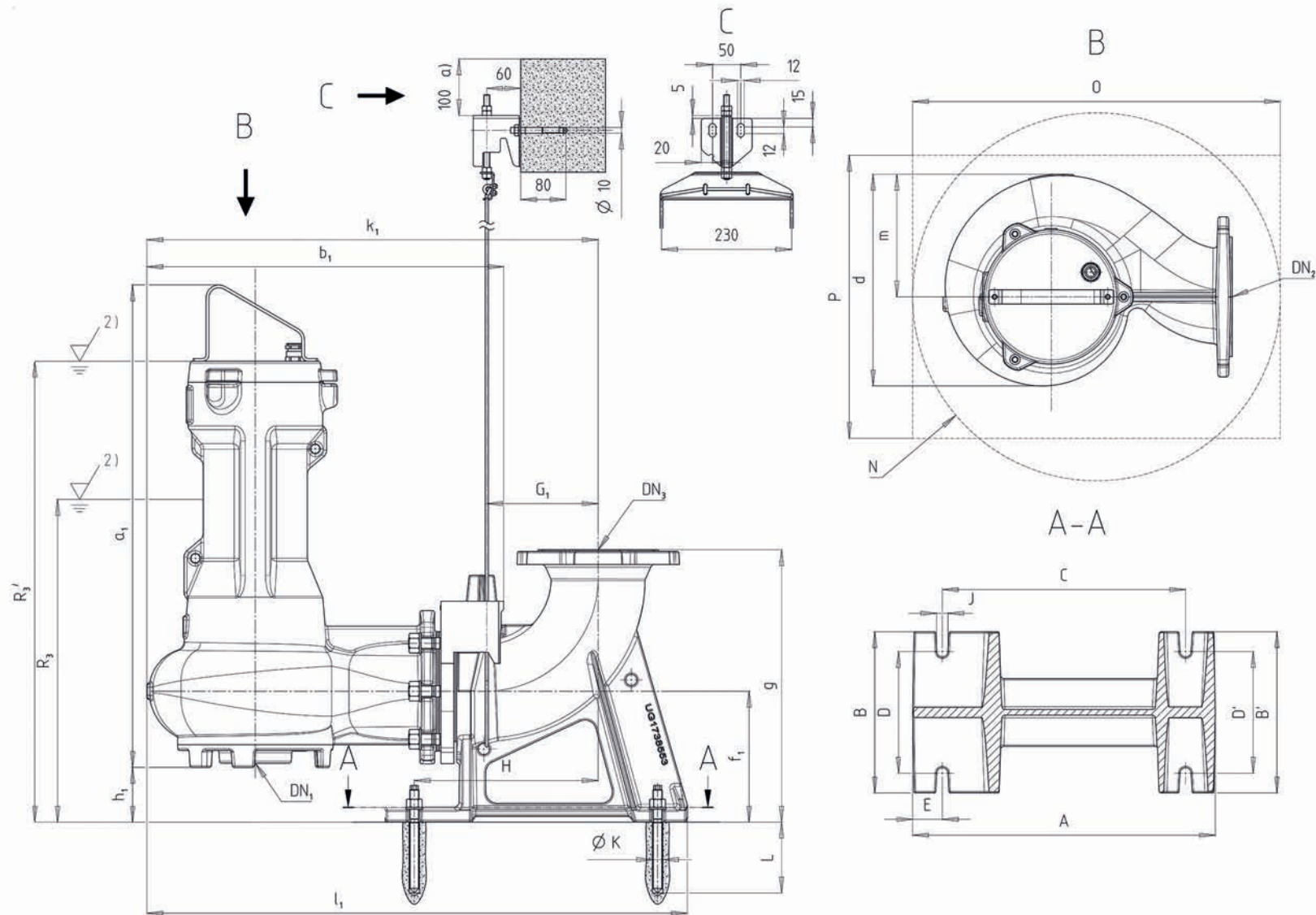


Fig. 10: Amarex 150, stationary installation, guide wire arrangement (UG1752531)

1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

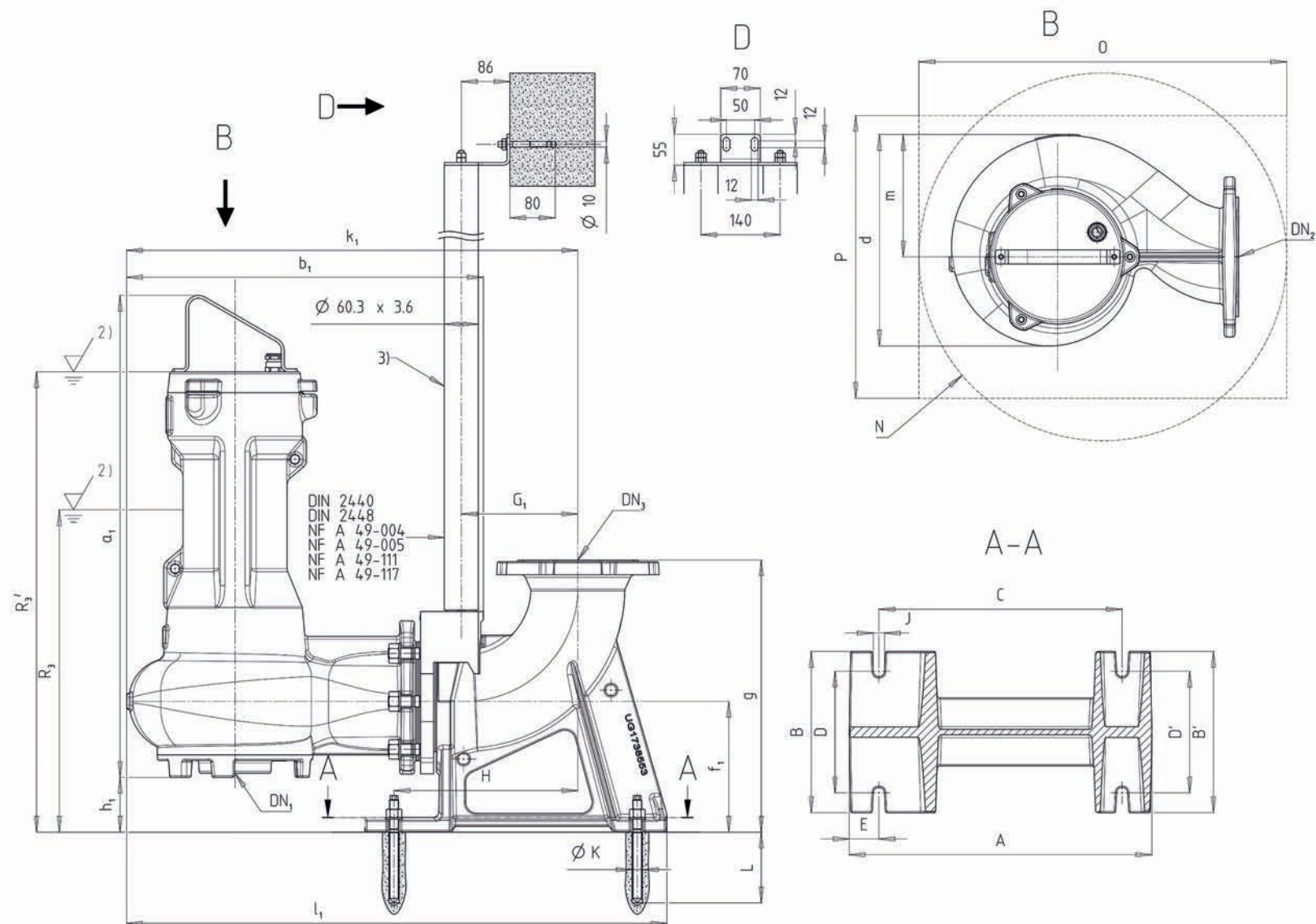


Fig. 11: Amarex 150, stationary installation, twin guide rail arrangement (UG1752531)

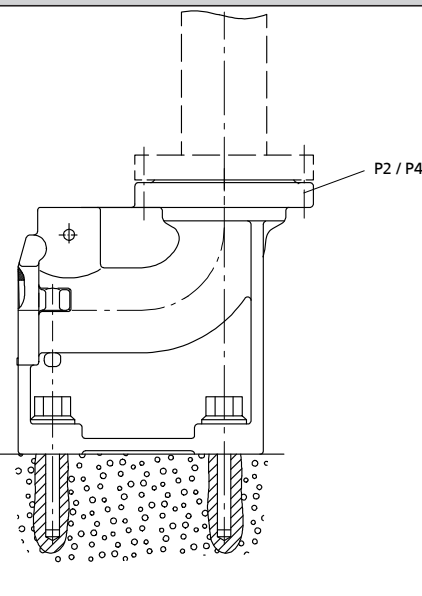
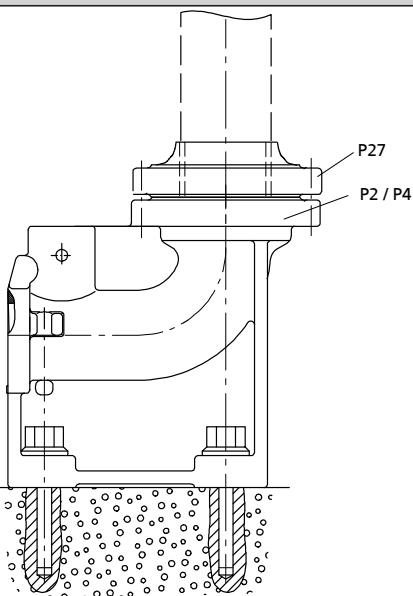
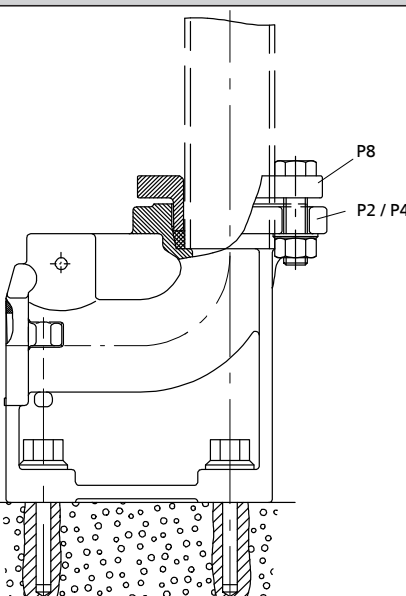
1) Lowest stop level for automatic operation 2) Minimum submergence for continuous duty 3) Not included in KSB's scope of supply a) Minimum b) Maximum

Dimensions of pump set and foundation [mm]

Amarex	Motor	Efficiency class	Pump																		Foundation (installation in a sump)																			
			DN ₁	DN ₂	a ₁		b ₁	d	D _f	f ₁	g	g _f	h ₁	H _f	k ₁	k _f	l ₁	m	Z _f	R ₃		R ₃ '		DN ₃	A	B	B'	C	D	D'	E	G ₁	G ₂	H	J	ØK	L	N	O	P
					US	YS														US	YS	US	YS															min.	min.	min.
F150-180	036 4	C	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	456	465	-	-	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
F150-180	065 4	F	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	-	-	806	815	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
D150-230	036 4	C	135	150	846	855	630	374	271	232	481	211	95	285	796	240	953	216	6	456	465	-	-	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
F150-230	036 4	C	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	456	465	-	-	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
D150-230	045 4	C	135	150	846	855	630	374	271	232	481	211	95	285	796	240	953	216	6	536	545	-	-	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
F150-230	045 4	C	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	536	545	-	-	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
D150-230	065 4	F	135	150	846	855	630	374	271	232	481	211	95	285	796	240	953	216	6	-	-	806	815	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
F150-230	065 4	F	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	-	-	806	815	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
D150-230	077 4	F	135	150	846	855	630	374	271	232	481	211	95	285	796	240	953	216	6	-	-	806	815	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500
F150-230	077 4	F	110	150	844	853	630	374	271	232	481	211	97	285	796	240	953	216	6	-	-	806	815	150	535	285	285	430	215	215	53	260	-	325	20	18	125	730	730	500

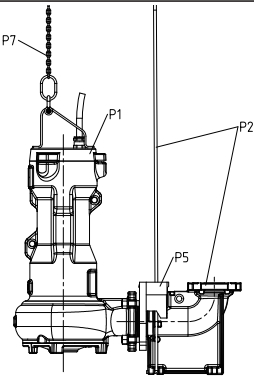
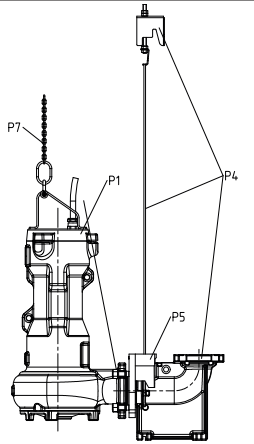
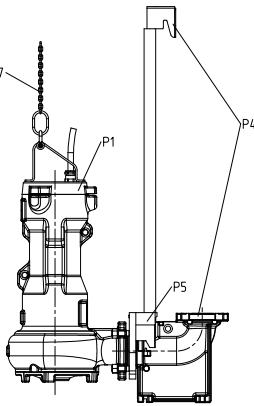
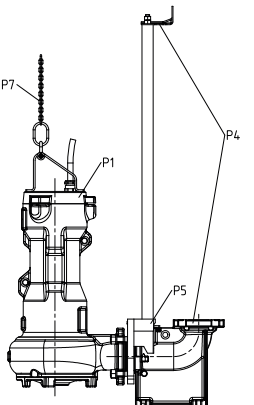
Connections

Discharge line connections at the duckfoot bend

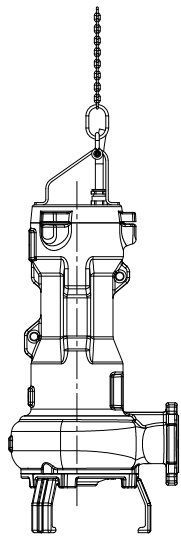
Flanged connection (DN 50/DN 65)	Threaded connection in the flange (DN 50/DN 65)	Clamped connection (DN 50/DN 65)
		
	For standard pipes to DIN 2440 / DIN 2441 ▪ With outside diameter of 60.3 mm, steel for DN 50 ▪ With outside diameter of 63 mm, PVC (ISO 3606) for DN 50 ▪ With threaded flange DN 50 - G 2" ▪ With outside diameter of 76.1 mm, steel for DN 65 ▪ With outside diameter of 75 mm, PVC (ISO 3606) for DN 65 ▪ With threaded flange DN 65 - G 2 1/2"	For standard pipes to DIN 2440 / DIN 2441 / DIN 2448 ▪ With outside diameter of 60.3 mm, steel for DN 50 ▪ With outside diameter of 63 mm, PVC (ISO 3606) for DN 50 ▪ With outside diameter of 76.1 mm, steel for DN 65 ▪ With outside diameter of 75 mm, PVC (ISO 3606) for DN 65

Installation types

Installation type S, stationary wet installation

Installation type	Description	Comment
	Guide hoop arrangement P1: pump P2: installation parts for guide hoop arrangement, installation depth = 1.5 m / 1.8 m / 2.1 m P5: claw P7: chain and shackle, length = 2 m	Only available for specific sizes, see selection configurator.
	Guide wire arrangement P1: pump P4: installation parts for guide wire arrangement, installation depth = 4.5 m P5: claw P7: chain and shackle, length = 5 m	
	Single guide rail arrangement P1: pump P4: installation parts for single guide rail arrangement P5: claw P7: chain and shackle, length = 5 m	Only available for specific sizes, see selection configurator.
	Twin guide rail arrangement P1: pump P4: installation parts for twin guide rail arrangement P5: claw and adapter P7: chain and shackle, length = 5 m	

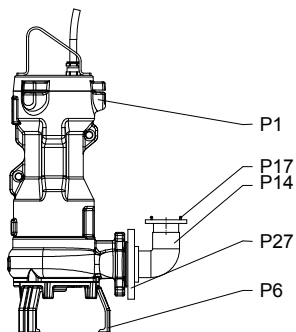
Installation type P, transportable wet-installed model

Installation type	Description
	P1: pump P6: pump foot P7: chain and shackle, length = 5 m

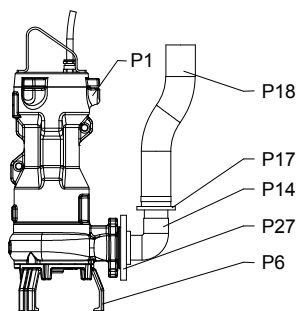
Installation information

Suggested installation layouts for transportable pump sets

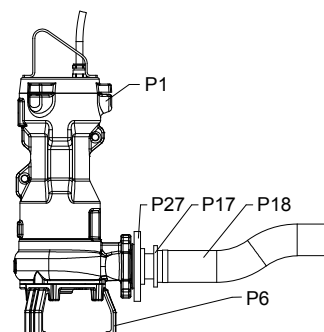
Size 50



Suggested installation layout 1
Vertical hose connection with elbow (P14) and Storz rigid coupling (P17) (quick connection)

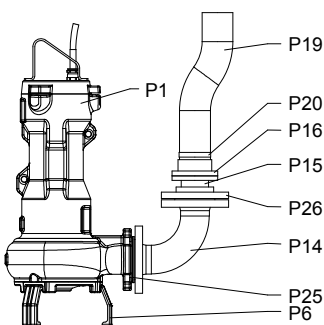


Suggested installation layout 2
Vertical hose connection with plastic hose (P18)

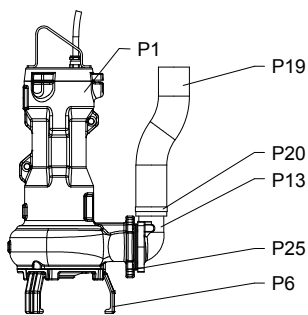


Suggested installation layout 3
Horizontal hose connection (quick connection) with plastic hose (P18)

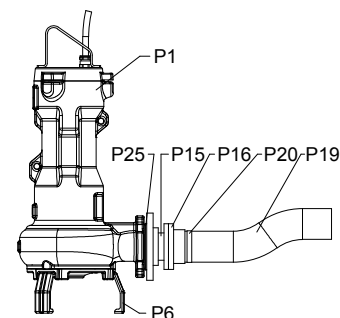
Sizes 65, 80, 100



Suggested installation layout 1
Vertical hose connection (quick connection) with plastic hose (P19) and hose clip (P20)



Suggested installation layout 2
Vertical hose connection with plastic hose (P19), hose clip (P20) and connection elbow (P13)



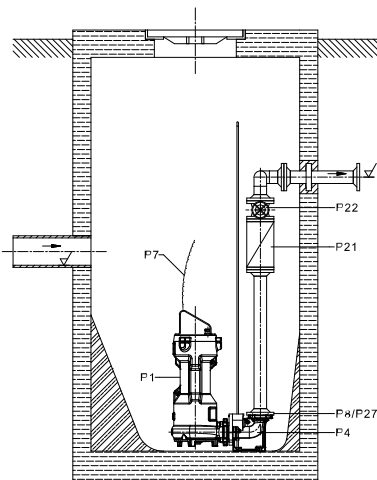
Suggested installation layout 3
Horizontal hose connection (quick connection) with plastic hose (P19), hose clip (P20), Storz hose coupling (P16), Storz rigid coupling (P15)

Suggested installation layouts for stationary pump sets

Sizes 50, 65, 80, 100

Guide hoop arrangement

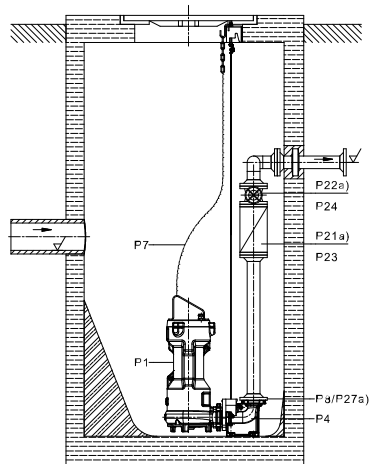
Amarex S 50-172/F 50-170, S 50-222/F 50-220, F 65-170/65-220



Suggested installation layout 1
Single-pump station
Flanged duckfoot bend

Guide wire, single guide rail or twin guide rail arrangement

Amarex 50, 65, 80, 100

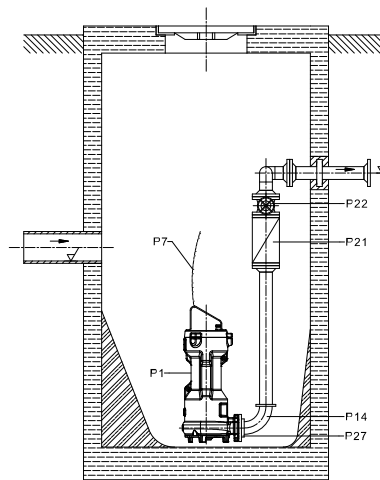


Suggested installation layout 3
Either with guide wire, single guide rail or twin guide rail
arrangement
Single-pump station for 4.5 m installation depth
Flanged duckfoot bend

a)	DN 50 only
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Suspended installation

Amarex S 50-172/F 50-170, S 50-222/F 50-220

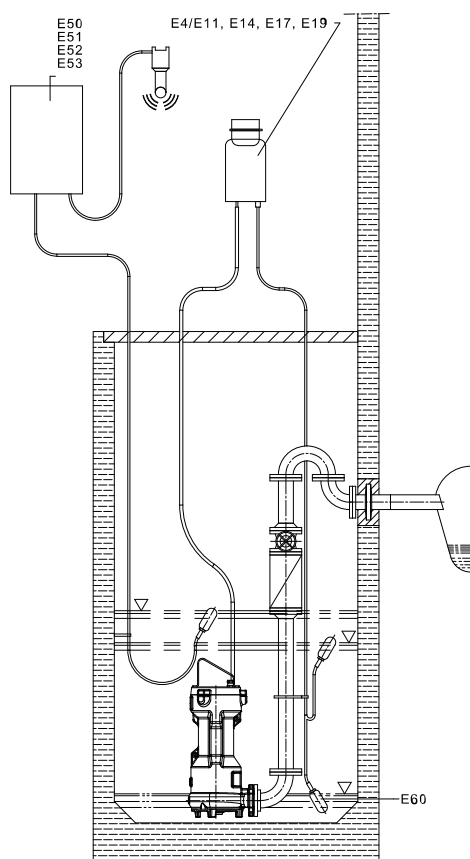


Suggested installation layout 2
Single-pump station for 1.5 m installation depth
Direct connection to discharge line (suspended installation)

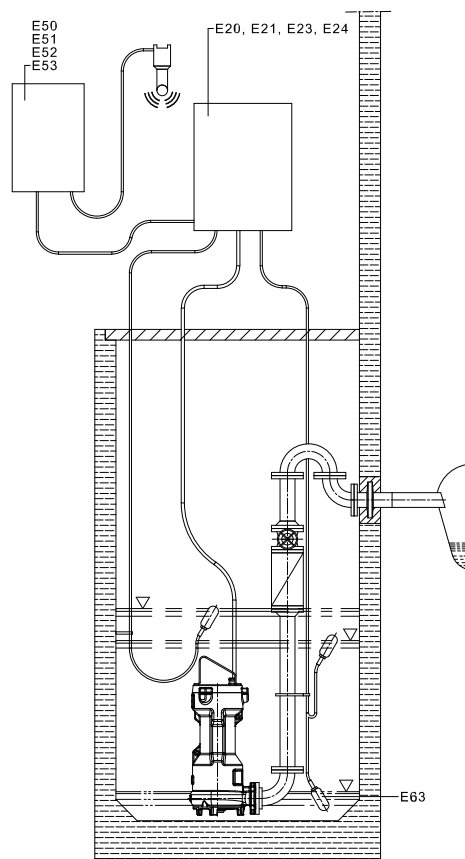
Suggested electrical installation layouts

Amarex is available in explosion-proof and non-explosionproof design.

CEE motor protection switch (up to 4.0 kW)



Suggested installation layout 1
without explosion protection



Suggested installation layout 2
with explosion protection

Scope of supply

Stationary wet-installed model (installation type S)

- Pump set, complete with power cable
- Claw with sealing elements and fasteners
- Lifting rope / lifting chain¹⁵⁾
- Mounting bracket with fasteners
- Duckfoot bend with mounting elements
- Guiding equipment¹⁶⁾

Transportable wet-installed model (installation type P)

- Pump set, complete with power cable
- Feet (and foot plate, if applicable)
- Lifting rope / lifting chain¹⁷⁾

15) Optional

16) The guide rails are not included in the scope of supply.

17) Optional

Accessories
Installation parts for stationary pump sets

Overview of installation parts for stationary pump sets

	Item	Description	Size	Sealing element	Installation depth [m]	Mat. No.	[kg]
	P2 + P5	Guide hoop arrangement Installation parts for stationary well installation Consisting of: DN 50 flanged duckfoot bend, stainless steel anchor bolts, guide hoop, claw with stainless steel screws/bolts	DN 50 DN 3: DIN ISO ASME Inclined claw	NBR 60	1,5	05045152	12,218
						19521154	1,5
				FKM 60	1,5	05045153	12,218
						19521154	1,5
				NBR 60	1,8	05045152	12,218
						19521155	2,5
				FKM 60	1,8	05045153	12,218
						19521155	2,5
		Guide hoop arrangement Installation parts for stationary well installation Consisting of: DN 50 flanged duckfoot bend, stainless steel anchor bolts, guide hoop, claw with stainless steel screws/bolts	DN 50 DN 3: DIN ISO ASME Inclined claw	NBR 60	2,1	05045152	12,218
						19521156	4
				FKM 60	2,1	05045153	12,218
						19521156	4
		Guide hoop arrangement Installation parts for stationary well installation Consisting of: DN 65 flanged duckfoot bend, stainless steel anchor bolts, guide hoop, claw with stainless steel screws/bolts	DN 65 DN 3: DIN ISO ASME	NBR 60	1,5	05045148	16,23
						19521154	1,5
				FKM 60	1,5	05045149	16,23
						19521154	1,5
				NBR 60	1,8	05045148	16,23
						19521155	2,5
				FKM 60	1,8	05045149	16,23
						19521155	2,5
		Guide hoop arrangement Installation parts for stationary well installation Consisting of: DN 65/80 flanged duckfoot bend, stainless steel anchor bolts, guide hoop, claw with stainless steel screws/bolts	DN 65/80 DN 3: DIN ISO	NBR 60	1,5	05045160	18,33
						19521154	1,5
				FKM 60	1,5	05045161	18,33
						19521154	1,5
				NBR 60	1,8	05045160	18,33
						19521155	2,5
				FKM 60	1,8	05045161	18,33
						19521155	2,5
			DN 65/80 DN 3: ASME	NBR 60	2,1	05045160	18,33
						19521156	4
				FKM 60	2,1	05045161	18,33
						19521156	4
				NBR 60	1,5	05045156	18,43
						19521154	1,5
				FKM 60	1,5	05045157	18,43
						19521154	1,5
				NBR 60	1,8	05045156	18,43
						19521155	2,5
				FKM 60	1,8	05045157	18,43
						19521155	2,5
				NBR 60	2,1	05045156	18,43
						19521156	4
				FKM 60	2,1	05045157	18,43
						19521156	4

2573.5/01-EN

	Item	Description	Size	Sealing element	Installation depth [m]	Mat. No.	[kg]
	P4 + P5	Guide wire arrangement Installation parts for stationary well installation Consisting of: flanged duckfoot bend, stainless steel anchor bolts, suspension bracket, mounting bracket, 10 m guide wire, claw with stainless steel screws/bolts	DN 50	NBR 60	4,5	05044127	13,85
			DN 3: DIN ISO ASME	FKM 60	4,5	05044128	13,85
			DN 65	NBR 60	4,5	05044125	17,862
			DN 3: DIN ISO ASME	FKM 60	4,5	05044126	17,862
			DN 65/80	NBR 60	4,5	05044211	19,962
			DN 3: DIN ISO	FKM 60	4,5	05044212	19,962
			DN 65/80	NBR 60	4,5	05044129	20,062
			DN 3: ASME	FKM 60	4,5	05044210	20,062
			DN 80	NBR 60	4,5	05018552	29,454
			DN 3: DIN ISO	FKM 60	4,5	05018553	29,454
			DN 80	NBR 60	4,5	05028880	27,404
			DN 3: ASME	FKM 60	4,5	05028881	27,404
			DN 80/100	NBR 60	4,5	05018556	28,854
			DN 3: DIN ISO ASME	FKM 60	4,5	05018557	28,854
			DN 100	NBR 60	4,5	05018554	34,633
			DN 3: DIN ISO ASME	FKM 60	4,5	05018555	34,633
	P4 + P5	Single guide rail arrangement Installation parts for stationary well installation Consisting of: flanged duckfoot bend, stainless steel anchor bolts, mounting bracket, claw with stainless steel screws/bolts (guide rail not included in KSB's scope of supply)	DN 50	NBR 60	4,5	05045152	12,218
			DN 3: DIN ISO ASME			05017151	0,32
				FKM 60	4,5	05045153	12,218
						05017151	0,32
			DN 65	NBR 60	4,5	05045148	16,23
			DN 3: DIN ISO ASME			05017151	0,32
				FKM 60	4,5	05045149	16,23
						05017151	0,32
			DN 65/80	NBR 60	4,5	05045160	18,33
			DN 3: DIN ISO			05017151	0,32
				FKM 60	4,5	05045161	18,33
						05017151	0,32
			DN 65/80	NBR 60	4,5	05045156	18,43
			DN 3: ASME			05017151	0,32
				FKM 60	4,5	05045157	18,43
						05017151	0,32
			DN 80	NBR 60	4,5	05018726	27,892
			DN 3: DIN ISO			05017151	0,32
				FKM 60	4,5	05018727	27,892
						05017151	0,32
			DN 80	NBR 60	4,5	05028255	25,892
			DN 3: ASME			05017151	0,32
				FKM 60	4,5	05028256	25,892
						05017151	0,32
			DN 80/100	NBR 60	4,5	05018742	27,292
			DN 3: DIN ISO ASME			05017151	0,32
				FKM 60	4,5	05018743	27,292
						05017151	0,32
			DN 100	NBR 60	4,5	05018729	33,071
			DN 3: DIN ISO ASME			05017151	0,32
				FKM 60	4,5	05018740	33,071
						05017151	0,32

	Item	Description	Size	Sealing element	Installation depth [m]	Mat. No.	[kg]
	P4 + P5	Twin guide rail arrangement Installation parts for stationary well installation Consisting of: flanged duckfoot bend, stainless steel anchor bolts, mounting bracket, adapter, claw with stainless steel screws/bolts (guide rails not included in KSB's scope of supply)	DN 50	NBR 60	-	05044215	13,33
			DN 3: DIN ISO ASME	FKM 60	-	05044216	13,33
			DN 65	NBR 60	-	05044213	17,242
			DN 3: DIN ISO ASME	FKM 60	-	05044214	17,242
			DN 65/80	NBR 60	-	05044219	19,342
			DN 3: DIN ISO	FKM 60	-	05044220	19,342
			DN 65/80	NBR 60	-	05044217	19,442
			DN 3: ASME	FKM 60	-	05044218	19,442
			DN 80	NBR 60	-	05018645	30,244
			DN 3: DIN ISO	FKM 60	-	05018646	30,244
			DN 80	NBR 60	-	05028447	28,194
			DN 3: ASME	FKM 60	-	05028448	28,194
			DN 80/100	NBR 60	-	05018649	29,644
			DN 3: DIN ISO ASME	FKM 60	-	05018650	29,644
			DN 100	NBR 60	-	05018647	35,423
			DN 3: DIN ISO ASME	FKM 60	-	05018648	35,423
DN 150	NBR 60	-	05016943	73,25			
DN 3: DIN ISO ASME	FKM 60	-	05016944	73,25			
	P5	Guide hoop arrangement, guide wire arrangement, single guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 50	NBR 60	-	05046255	4,842
				FKM 60	-	05046256	4,842
		Single guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 65	NBR 60	-	05046259	5,854
				FKM 60	-	05046270	5,854
		Single guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 80	NBR 60	-	05016790	6,572
				FKM 60	-	05016791	6,572
		Single guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 100	NBR 60	-	05016794	7,291
				FKM 60	-	05016795	7,291
		Twin guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 50	NBR 60	-	05046257	4,942
				FKM 60	-	05046258	4,942
		Twin guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 65	NBR 60	-	05046271	5,854
				FKM 60	-	05046272	5,854
		Guide wire arrangement, twin guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 80	NBR 60	-	05016792	6,372
				FKM 60	-	05016793	6,372
		Guide wire arrangement, twin guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 100	NBR 60	-	05016796	7,091
				FKM 60	-	05016797	7,091
Guide wire arrangement, twin guide rail arrangement Claw for Amarex EN-GJL-250 with stainless steel screws/bolts	DN 150	NBR 60	-	05016798	13,864		
		FKM 60	-	05016799	13,864		

Installation parts transportable pump sets

Overview of installation parts for transportable pump sets

	Item	Description	Size	Mat. No.	[kg]
	P6	3 pump feet	DN 50/65/80/100	05017148	0,072
		Footplate with bolts (only for uneven mounting surfaces and in combination with feet)	DN 50/65/80/100	05016969	1,004

Chain for stationary and transportable pump sets

Overview of chains for stationary and transportable pump sets

	Item	Description	Amarex	Length	Mat. No.	[kg]
				[m]		
	P7	Chain (1.4404) short-linked, tested and duly labelled to Directive 2006/42/EC (Machinery Directive), hook (1.4301), shackle (1.4404), maximum load 200 kg	G08-09	3	05018486	0,793
			G10-11	3	05018487	0,943
			G08-09	5	05018488	2,295
			G10-11	5	05018489	2,445
			G08-09	10	05018510	4,4
			G10-11	10	05018511	4,55
			G08-09	15	05018512	6,605
			G10-11	15	05018513	6,755
	-	Polypropylene lifting rope with shackle 1.4401 and hook 1.4571	G08-09	5	05018515	2,09
			G10-11	5	05018517	2,24

Pump accessories

Overview of accessories for stationary and transportable pump sets

	Item	Description	Connection	Length	Amarex				Mat. No.	[kg]
				[m]	50	65	80	100		
	P8	Flange for pipe coupling PN 10, at the flanged elbow, mating dimensions to PN 16	DN 50 / R 2	-	X	-	-	-	19551111	1,2
			DN 65 / R 2 1/2	-	-	X	-	-	39020184	1,2
	P9	PVC adapter for hose connection, with 1 hose clip	R2	-	X	-	-	-	11191498	0,3
	P13	Connection elbow with flange/hose connection made of EN-GJL-250, grey cast iron PN 16, DIN 2501, including joint ring and hose clip; for DN 100 also with fixing bolts, To be used for flange connections item 25 / item 26 (not for DN 100).	DN 65/B 75	-	-	X	-	-	19135655	6
			DN 80/B 75	-	-	-	X	-	19131746	6,6
			DN 100/A 110	-	-	-	-	X	19139718	10
	P14	Angle with internal thread / external thread, galvanised grey cast iron (flange connection see P27 and P30)	R 2	-	X	-	-	-	00241966	0,3
		Connection elbow with flanges PN 16, DIN 2501 (to be used for flange connections item 25 or item 26), grey cast iron	DN 65/65	-	-	X	-	-	00265480	11
			DN 65/80	-	-	X	-	-	25198402	8
			DN 80/80	-	-	-	X	-	11150856	10
			DN 100/100	-	-	-	-	X	25145802	14,4
	P15	Storz rigid coupling	DN 65 / B 75	-	-	X	-	-	18040148	3,5
			DN 80/B 75	-	-	-	X	-	18072642	3,5
			DN 100/A 110	-	-	-	-	X	18060162	5
	P16	Storz hose coupling 2 hose clips, item 20, are required for hose mounting (for plastic hoses B 75 and A 110, item 19)	C 52 (DIN 14321)	-	X	-	-	-	00524551	0,3
			B 75 (DIN 14322)	-	-	X	X	-	00520454	0,7
			A 110 (DIN 14323)	-	-	-	-	X	00522313	1,5
	P17	Storz rigid coupling	C 52 / G 2	-	X	-	-	-	00524370	0,2
			B 75 / G 2 1/2	-	-	X	-	-	00524371	0,4
	P18	Plastic hose DN 50, DIN 14811, with integrated C couplings	C 52	-	X	-	-	-	00522262	2,3
			C 52	-	X	-	-	-	00522263	4,2
			C 52	-	X	-	-	-	00522264	5,7
	P19	Plastic hose, without coupling, DIN 14811 Inside diameter 63 mm	-	5	X	-	-	-	39018688	1,7
			-	10	X	-	-	-	39018689	3,4
			-	20	X	-	-	-	39018690	6,8
		Plastic hose without coupling, DIN 14 811 B 75	-	5	-	X	X	-	39019064	2
			-	20	-	X	X	-	39019066	8
			-	30	-	X	X	-	39019071	12

	Item	Description	Connection	Length	Amarex				Mat. No.	[kg]
				[m]	50	65	80	100		
	P19	Plastic hose without coupling, DIN 14 811 80	-	5	-	-	X	-	39018691	2,2
			-	10	-	-	X	-	39019062	4,3
		Plastic hose without coupling, DIN 14 811 A 110	-	5	-	-	-	X	39019067	4,7
			-	10	-	-	-	X	39019068	9,3
			-	30	-	-	-	X	39019070	27,9
	P20	Hose clip DIN 3017, chrome steel	B 50 ¹⁸⁾	-	X	-	-	-	39000515	0,025
			B 75	-	-	X	X	-	00109515	0,04
			AL 110 - 120 B ¹⁹⁾	-	-	-	-	X	00520853	0,1
	P21	RK swing check valve Plastic, EN 12050-4, with internal thread ISO 7/1, full port and drain plug; cannot be used for pumped drainage	Rp 2	-	X	-	-	-	01009773	0,5
	P22	Socket gate valve, CuZn, PN 10-12 DIN 3352	Rp 2	-	X	-	-	-	00411503	1,287
			Rp 2 1/2	-	-	X	-	-	39000507	1,7
	P23	Check valve, Grey cast iron, full bore, lifting device, flanges drilled to DIN 2501, PN 16	DN 65	-	-	X	-	-	48829253	13,74
			DN 80	-	-	-	X	-	48829254	16,5
			DN 100	-	-	-	-	X	48829255	20,9
	P24	ECOLINE GTR-16P gate valve, grey cast iron, PN 16, flanges drilled to ISO 7005/DIN 2501	DN 65	-	-	X	-	-	49709579	15
			DN 80	-	-	-	X	-	49709580	22
			DN 100	-	-	-	-	X	49709581	26,5
	P25	Set of installation accessories for a flange connection, discharge nozzle; item 13, 14 or 15 Consisting of: 4 hexagon head bolts with nuts and 1 sealing element	-	-	X	-	-	-	39021944	0,8
					-	X	-	-	19551115	0,8
					-	-	X	-	19551100	0,8
					-	-	-	X	19551113	0,8
	P26	Set of installation accessories for a flange connection Consisting of: 8 hexagon head bolts with nuts and 1 sealing element	-	-	-	-	X	-	19551114	0,8
					-	-	-	X	19551116	0,8
	P27	Threaded flange for guide hoop arrangement (P2) and guide wire arrangement (P4) Consisting of: flange, 4 hexagon head bolts with nuts and washers and 1 sealing element	DN 50 / Rp 2	-	X	-	-	-	19551353	2
			DN 65 / Rp 2 1/2	-	-	X	-	-	39021943	2,9
	P28	Hand pump, wall mounting, grey cast iron, suction-side connection Rp 1 1/2	-	-	X	X	X	X	00520485	12
	P30	Adapter with reduced external thread M4 EN 1042K	2 x Rp 1 1/4	-	-	-	-	-	01135663	0,4

18) For plastic hose Ø 63 item 19
19) 2 units required

Non-ATEX-compliant control units / switchgear

Overview of control units and switchgear for Amarex, non-ATEX-compliant ( Valid for all countries except France.)

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
	E4	Multi-functional plug, type Hyper, with motor protection relay CEE plug	2,6	3,7	Hyper 37.1	19071492	1
			3,7	5,5	Hyper 55.1	19071493	1
			5,5	8,0	Hyper 80.1	19071494	1
			8,0	11,5	Hyper 115.1	19071495	1
LevelControl Basic 2 control unit for single-pump station, IP54							
	E11	For float switch or 4 - 20 mA sensor, optionally with master switch, 400 x 281 x 135 mm	2,5	4,0	BC1 400 DFNO 040	19073763	4,5
			4,0	6,3	BC1 400 DFNO 063	19073764	4,5
			6,3	10,0	BC1 400 DFNO 100	19073765	4,5
	E14	Pneumatic, optional master switch, 400 x 281 x 135 mm	2,5	4,0	BC1 400 DPNO 040	19073768	4,5
			4,0	6,3	BC1 400 DPNO 063	19073769	4,5
			6,3	10,0	BC1 400 DPNO 100	19073770	4,5
	E17	Bubbler control with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 DLNO 040	19073818	12
			4,0	6,3	BS1 400 DLNO 063	19073819	12
			6,3	10,0	BS1 400 DLNO 100	19073820	12
	E19	Bubbler control in BC version Only to be used for connections with a neutral conductor. Optional component O1 (master switch) not possible. 400 x 281 x 135 mm	2,5	4,0	BC1 400 DLNO 040	19075148	4,5
			4,0	6,3	BC1 400 DLNO 063	19075149	4,5
			6,3	10,0	BC1 400 DLNO 100	19075150	4,5
LevelControl Basic 2 control unit for dual-pump station, IP54							
	E31	For float switch or 4 - 20 mA sensor, optionally with master switch, 400 x 281 x 135 mm	2,5	4,0	BC2 400 DFNO 040	19073777	4,7
			4,0	6,3	BC2 400 DFNO 063	19073778	4,7
			6,3	10,0	BC2 400 DFNO 100	19073779	4,7
	E34	Pneumatic, optional master switch, 400 x 281 x 135 mm	2,5	4,0	BC2 400 DPNO 040	19073782	4,7
			4,0	6,3	BC2 400 DPNO 063	19073783	4,7
			6,3	10,0	BC2 400 DPNO 100	19073784	4,7
	E37	Bubbler control with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 DLNO 040	19073860	13
			4,0	6,3	BS2 400 DLNO 063	19073861	13
			6,3	10,0	BS2 400 DLNO 100	19073862	13
	E39	Bubbler control in BC version Only to be used for connections with a neutral conductor. Optional component O1 (master switch) not possible. 400 x 281 x 135 mm	2,5	4,0	BC2 400 DLNO 040	19075151	4,7
			4,0	6,3	BC2 400 DLNO 063	19075152	4,7
			6,3	10,0	BC2 400 DLNO 100	19075153	4,7
LevelControl Basic 2 optional components (processed via KSB EasySelect) ²⁰⁾							
	O1	Master switch for LevelControl Basic 2 BC, fitted	-	-	-	01143084	0,2
	O2	Control cabinet heating for type BS, fitted with 20 W thermostat	-	-	-	19074269	0,3
	O10	Outdoor cabinet type 142 with base, for type BC IP 44, glass fibre reinforced polyester, colour: RAL 7035, locking device: profile half cylinder, can be buried Internal dimensions H x W x D [mm]: 600 x 276 x 165	-	-	-	19071911	15
	O11	Outdoor cabinet type 0/845 for types BS1 (up to 25 A) and BS2 (up to 10 A) IP44, glass fibre reinforced polyester, colour: RAL 7035, DIN 43 629, locking device: profile half cylinder, can be buried, including metal frame for setting in concrete Dimensions H x W x D [mm]: 585 x 845 x 315	-	-	-	19071440	40
	O200	Signalling module for LevelControl Basic 2 BC	-	-	-	19075182	0,2
	O201	Signalling module for type BC, with pressure sensor 3 mWC for redundant pneumatic level measurement or redundant bubbler system	-	-	-	19075183	1,1
	O203	Signalling module for type BS	-	-	-	19075185	1,1

20) Optional components must be processed via KSB EasySelect, otherwise they will be supplied but not fitted.

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
	O204	Signalling module for type BS, with pressure sensor 3 mWC for redundant pneumatic level measurement or redundant bubbler system	-	-	-	19075186	0,8

Overview of control units and switchgear for Amarex, non-ATEX-compliant (i Only valid for France.)

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
	E4	Multi-functional plug, type Hyper, with motor protection relay CEE plug	2,6	3,7	Hyper 37.1	19071492	1
			3,7	5,5	Hyper 55.1	19071493	1
			5,5	8,0	Hyper 80.1	19071494	1
			8,0	11,5	Hyper 115.1	19071495	1
LevelControl Basic 2 control unit for single-pump station							
	E11	For float switch or 4 - 20 mA sensor, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 DFNO 040 02	19073878	4,5
			4,0	6,0	BC1 400 DFNO 063 02	19073879	4,5
			6,0	10,0	BC1 400 DFNO 100 02	19073880	4,5
	E14	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 DPNO 040 02	²¹⁾	²¹⁾
			4,0	6,3	BC1 400 DPNO 063 02	²¹⁾	²¹⁾
			6,3	10,0	BC1 400 DPNO 100 02	²¹⁾	²¹⁾
	E17	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 DLNO 040 02	²¹⁾	²¹⁾
			4,0	6,3	BS1 400 DLNO 063 02	²¹⁾	²¹⁾
			6,3	10,0	BS1 400 DLNO 100 02	²¹⁾	²¹⁾
LevelControl Basic 2 control unit for dual-pump station							
	E11	For float switch or 4 - 20 mA sensor, optional master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 DFNO 040 02	19073888	4,7
			4,0	6,3	BC2 400 DFNO 063 02	19073889	4,7
			6,0	10,0	BC2 400 DFNO 100 02	19073890	4,7
	E14	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 DPNO 040 02	²¹⁾	²¹⁾
			4,0	6,3	BC2 400 DPNO 063 02	²¹⁾	²¹⁾
			6,3	10,0	BC2 400 DPNO 100 02	²¹⁾	²¹⁾
	E17	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 DLNO 040 02	²¹⁾	²¹⁾
			4,0	6,3	BS2 400 DLNO 063 02	²¹⁾	²¹⁾
			6,3	10,0	BS2 400 DLNO 100 02	²¹⁾	²¹⁾

²¹⁾ See KSB Easy Select

Alarm switchgears for pumps, non-ATEX-compliant

AS 0/AS 1/AS 2/AS 4/AS 5

Item	Description	Mat. No.	[kg]
E50	 Alarm switchgear AS 0 with circuit breaker, acoustic signal transmitter with 85 dB(A), green equipment-on lamp Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64), M1 alarm contactor or signal relay of control unit as contactor.	29128401	0,5
E51	 Alarm switchgear AS 2 With circuit breaker, acoustic signalling device with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch, F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128422	0,5
E52	 Alarm switchgear AS 4 with circuit breaker, acoustic signal transmitter with 85 dB(A), green equipment-on lamp, volt-free contact for hook-up to a control station, self-charging power supply unit for 5 hours of operation in the event of a power failure Plastic housing, IP20, H x W x D = 140 x 80 x 57 [mm]. Use float switch (E60), F1 leakage sensor (item E64) or signal relay of control unit as contactor.	29128442	0,5
E53	 Alarm switchgear AS 5 Mains-independent, with self-charging power supply unit for 10 hours of operation in the event of a power failure, mains pilot LED, fault indicator light, horn-off pushbutton, volt-free contact for hook-up to a control station, ready for connection with 1.8 m power cable and plug. ISO housing, IP41, H x W x D = 190 x 165 x 75 [mm]. Use float switch (E60) or signal relay of control unit as contactor.	00530561	1,7
E55	 Alarm switchgear AS 1 In IP30 ISO plug housing, mains-independent, with self-charging power supply unit for 5 hours of operation in the event of a power failure, acoustic signal transmitter 70 dB(A) with circuit breaker and integrated signal transmitter with 3-meter power cable, max. 60 °C, not suitable for steam and condensate. 1. High water alert by suspending the sensor in a (pump) sump above the pump start-up point. 2. Water alarm signal at a water level of only 1 mm, by placing the sensor on the floor in areas with a flooding or leakage risk, e.g. the cellar or next to the washing machine in the kitchen or bathroom.	00533740	0,9

Accessories for non-ATEX-compliant control units

Overview of control unit accessories for Amarex, non-ATEX-compliant

	Item	Description	Length	Mat. No.	[kg]
			[m]		
	E60	Float switch with free cable end Function: circuit closed in upper float position (NO contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: H 07 RN-F3G1	3	11037742	0,5
			5	11037743	0,8
			10	11037744	1,3
			15	11037745	1,8
			20	11037746	2,4
			25	11037747	2,9
			30	11037748	3,4
	E62	Float switch with free cable end Function: circuit open in upper float position (NC contact) Float switch housing: polypropylene Fluid temperature: 70 °C max. Power cable: H 07 RN-F3G1	5	11037756	0,8
			10	11037757	1,4
			20	11037758	2,6
	E64	F1 leakage sensor Contactor for alarm switchgears AS 0, AS 2, AS 4 or as alarm transmitter for LevelControl Basic 2 Alarm transmission options: High water alert by suspending the sensor in a (pump) sump above the pump start-up point. Warning at a water level of 1 mm in areas with a flooding or leakage risk (e.g. in the cellar or next to the washing machine in the kitchen or bathroom) Dimensions [mm]: 52 x 21 x 20 (H x W x D)	3 m	19072366	0,2
	E65	Open pressure bell set, pneumatic and bubbler control with polyamide tube 8 x 1 mm	10	19071721	1,2
			20	19071837	2
			50	19074200	2,5
	E66	Closed pressure bell set, pneumatic with polyamide tube 8 x 3 mm	10	19071722	3,5
			> 10	On request	
	E70	Horn, 12 V DC, 105 dB, 150 mA, IP54	-	01086547	0,1
	E71	Alarm combination, 12 V DC	-	01139930	0,1
	E72	Yellow alarm strobe light, 12 V DC, 195 mA, IP65	-	01056355	0,3
	O45	Plastic housing, IP65, for easier installation of alarm strobe light, for wall mounting	-	01061067	0,2
	E73	KSB ServiceTool	-	47121210	0,2
	E90	Rechargeable battery retrofit kit for LevelControl Basic 2, BC Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charging unit	-	19074194	0,8
	E91	Rechargeable battery retrofit kit for LevelControl Basic 2, type BS Scope of supply: 1 rechargeable battery (12 V, 1.2 Ah) and charging unit	-	19074199	1

LevelControl Basic 2 control units, ATEX-compliant

i Control units for ATEX applications are not explosion-proof, which means that they must not be operated in potentially explosive atmospheres.

i For ATEX-compliant variants, intrinsic safety barriers and an ATEX level switch (float switch) must be fitted. Selection via KSB EasySelect.

Accessories for control units and switchgear for Amarex, ATEX-compliant (**i** **Valid for all countries except France!**)

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
LevelControl Basic 2 control unit for single-pump station							
	E20	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS1 400 DFEO 040	19073800	12
			4,0	6,3	BS1 400 DFEO 063	19073801	12
			6,3	10,0	BS1 400 DFEO 100	19073802	12
	E21	Pneumatic, optional master switch, 400 x 281 x 135 mm	2,5	4,0	BC1 400 DPEO 040	19073771	4,5
			4,0	6,3	BC1 400 DPEO 063	19073772	4,5
			6,3	10,0	BC1 400 DPEO 100	19073773	4,5
	E23	Bubbler control with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 DLEO 040	19073821	12
			4,0	6,3	BS1 400 DLEO 063	19073822	12
			6,3	10,0	BS1 400 DLEO 100	19073823	12
	E24	Bubbler control in BC version Only to be used for connections with a neutral conductor. Optional component O1 (master switch) not possible. 400 x 281 x 135 mm	2,5	4,0	BC1 400 DLEO 040	19075154	4,5
			4,0	6,3	BC1 400 DLEO 063	19075155	4,5
			6,3	10,0	BC1 400 DLEO 100	19075156	4,5
LevelControl Basic 2 control unit for dual-pump station							
	E40	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS2 400 DFEO 040	19073842	13
			4,0	6,3	BS2 400 DFEO 063	19073843	13
			6,3	10,0	BS2 400 DFEO 100	19073844	13
	E41	Pneumatic, optional master switch, 400 x 281 x 135 mm	2,5	4,0	BC2 400 DPEO 040	19073785	4,7
			4,0	6,3	BC2 400 DPEO 063	19073786	4,7
			6,3	10,0	BC2 400 DPEO 100	19073787	4,7
	E43	Bubbler control with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 DLEO 040	19073863	13
			4,0	6,3	BS2 400 DLEO 063	19073864	13
			6,3	10,0	BS2 400 DLEO 100	19073865	13
	E44	Bubbler control in BC version Only to be used for connections with a neutral conductor. Optional component O1 (master switch) not possible. 400 x 281 x 135 mm	2,5	4,0	BC2 400 DLEO 040	19075157	0,01
			4,0	6,3	BC2 400 DLEO 063	19075158	4,7
			6,3	10,0	BC2 400 DLEO 100	19075159	4,7
LevelControl Basic 2 optional components (processed via KSB EasySelect) ²²⁾							
	O7	Intrinsic safety barrier for additional float switch in potentially explosive atmospheres for example, high water float switch for pneumatic measurement or bubbler control in potentially explosive atmospheres Only in combination with type BS: Stahl 9002/13-280-093-001	-	-	-	01085568	0,2

Accessories for control units and switchgear for Amarex, ATEX-compliant (**i** **Only valid for France!**)

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
LevelControl Basic 2 control unit for single-pump station							
	E20	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS1 400 DFEO 040 02	23)	23)
			4,0	6,3	BS1 400 DFEO 063 02	23)	23)
			6,3	10,0	BS1 400 DFEO 100 02	23)	23)
	E21	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC1 400 DPEO 040 02	23)	23)
			4,0	6,3	BC1 400 DPEO 063 02	23)	23)
			6,3	10,0	BC1 400 DPEO 100 02	23)	23)

22) Optional components must be processed via KSB EasySelect, otherwise they will be supplied but not fitted.

23) See KSB EasySelect

	Item	Description	I _N	I _N	Type	Mat. No.	[kg]
			Min.	Max.			
			[A]	[A]			
	E23	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS1 400 DLEO 040 02	23)	23)
			4,0	6,3	BS1 400 DLEO 063 02	23)	23)
			6,3	10,0	BS1 400 DLEO 100 02	23)	23)
LevelControl Basic 2 control unit for dual-pump station							
	E40	Float switch, with master switch, 600 x 400 x 200 mm	2,5	4,0	BS2 400 DFEO 040 02	23)	23)
			4,0	6,3	BS2 400 DFEO 063 02	23)	23)
			6,3	10,0	BS2 400 DFEO 100 02	23)	23)
	41	Pneumatic, with master switch, 400 x 278 x 120 mm	2,5	4,0	BC2 400 DPEO 040 02	23)	23)
			4,0	6,3	BC2 400 DPEO 063 02	23)	23)
			6,3	10,0	BC2 400 DPEO 100 02	23)	23)
	E43	Bubbler control, with master switch, 400 x 300 x 155 mm	2,5	4,0	BS2 400 DLEO 040 02	23)	23)
			4,0	6,3	BS2 400 DLEO 063 02	23)	23)
			6,3	10,0	BS2 400 DLEO 100 02	23)	23)

Accessories for ATEX-compliant control units

Overview of accessories for Amarex with ATEX

	Item	Description	Length	Mat. No.	[kg]
			[m]		
	E63	Float switch with free cable end (NO contact) With declaration of compliance with explosion protection standards Power cable: (H07RN-F 3G1)	5	01148226	0,7
			10	01148247	1
			20	01148248	2
	E65	Open pressure bell set, pneumatic and bubbler control with polyamide tube 8 x 1 mm	10	19071721	1,2
			20	19071837	2
			50	19074200	2,5
	E66	Closed pressure bell set, pneumatic with polyamide tube 8 x 3 mm	10	19071722	3,5
			> 10	On request	
	E70	Horn, 12 V DC, 105 dB, 150 mA, IP54not explosion-proof	-	01086547	0,1
	E71	Alarm combination, 12 V DC , not explosion-proof	-	01139930	0,1
	E72	Yellow alarm strobe light, 12 V DC, 195 mA, IP65, not explosion-proof	-	01056355	0,3
	O45	Plastic housing, IP65, for easier installation of alarm strobe light, for wall mounting	-	01061067	0,2
	E73	KSB ServiceTool	-	47121210	0,2
	E90	Rechargeable battery retrofit kit for LevelControl Basic 2, BC Scope of supply: 2 rechargeable batteries (6 V, 1.3 Ah) and charging unit	-	19074194	0,8
	E91	Rechargeable battery retrofit kit for LevelControl Basic 2, type BS Scope of supply: 1 rechargeable battery (12 V, 1.2 Ah) and charging unit	-	19074199	1

General assembly drawings with list of components

General assembly drawings for version US

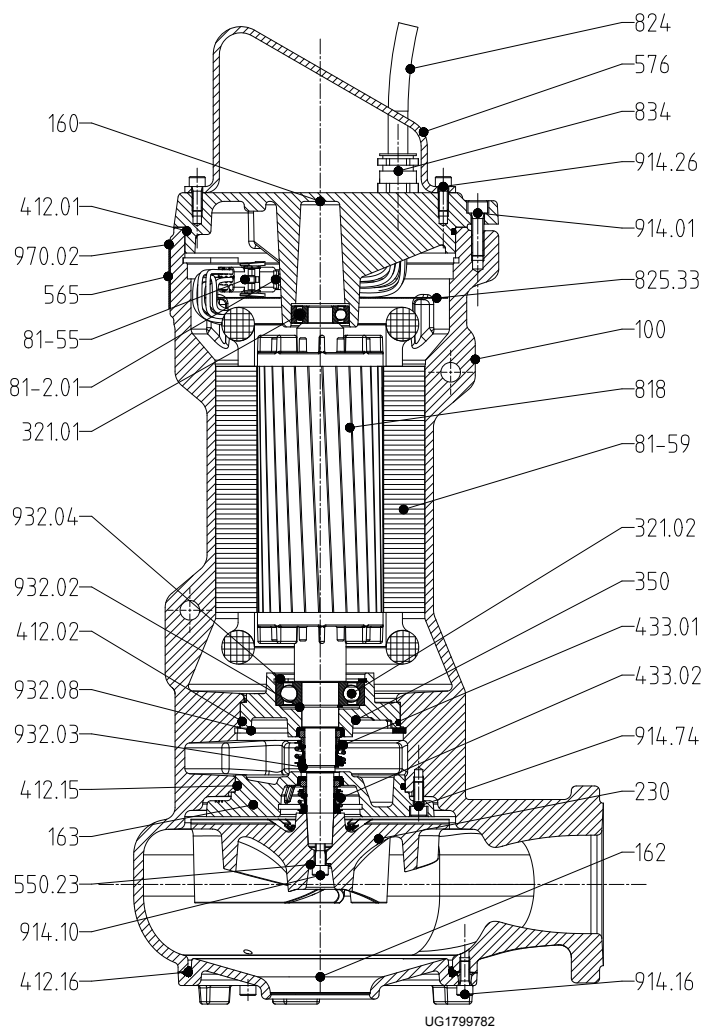


Fig. 12: General assembly drawing for version US, with impeller type F-max

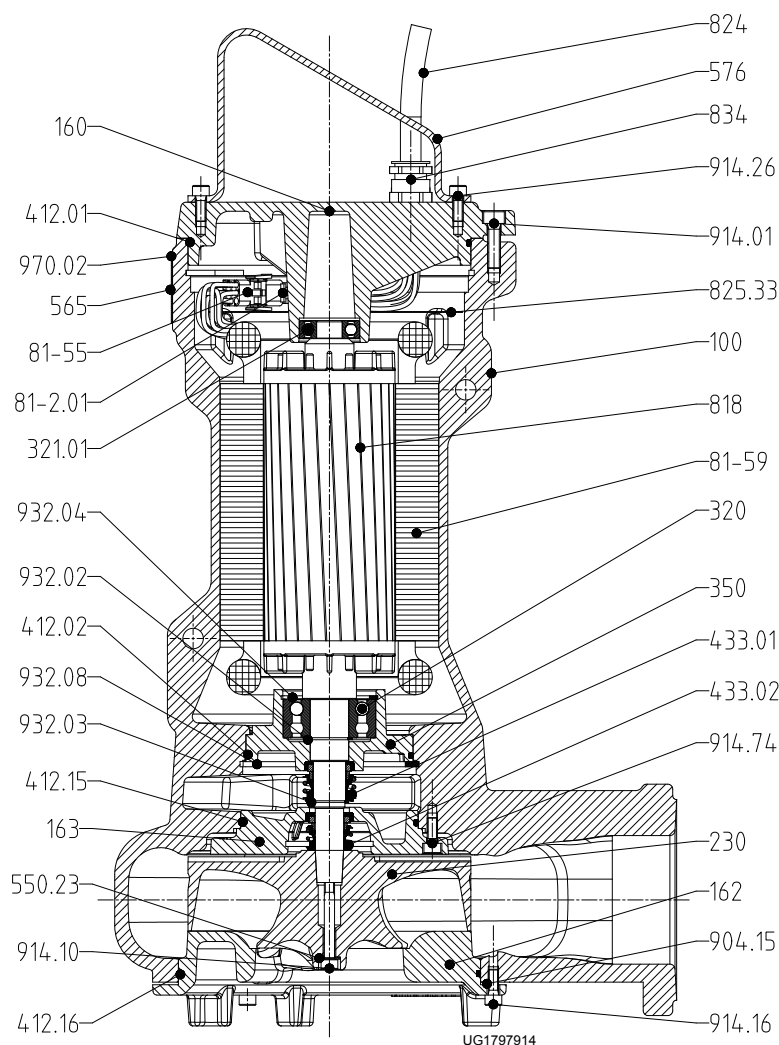


Fig. 13: General assembly drawing for version US, with impeller type D-max

List of components

Part No.	Description	Part No.	Description
100	Casing	576	Handle
160	Cover	81-2.01	Plug
162	Suction cover	81-55	Socket
163	Discharge cover	81-59	Stator
230	Impeller	818	Rotor
320 ²⁴⁾	Rolling element bearing	824	Cable
321.01/.02 ²⁵⁾	Radial ball bearing	825.33	Cable guard
350	Bearing housing	834	Cable gland
412.01/.02/.15/.16	O-ring	904.15 ²⁴⁾	Grub screw
433.01/.02	Mechanical seal	914.01/.10/.16/.26/.74	Hexagon socket head cap screw
550.23	Disc	932.02/.03/.04/.08	Circlip
565	Rivet	970.02	Label/plate

24) Only used for version with impeller type D-max

25) Only used for version with impeller type F-max

General assembly drawing for version YS

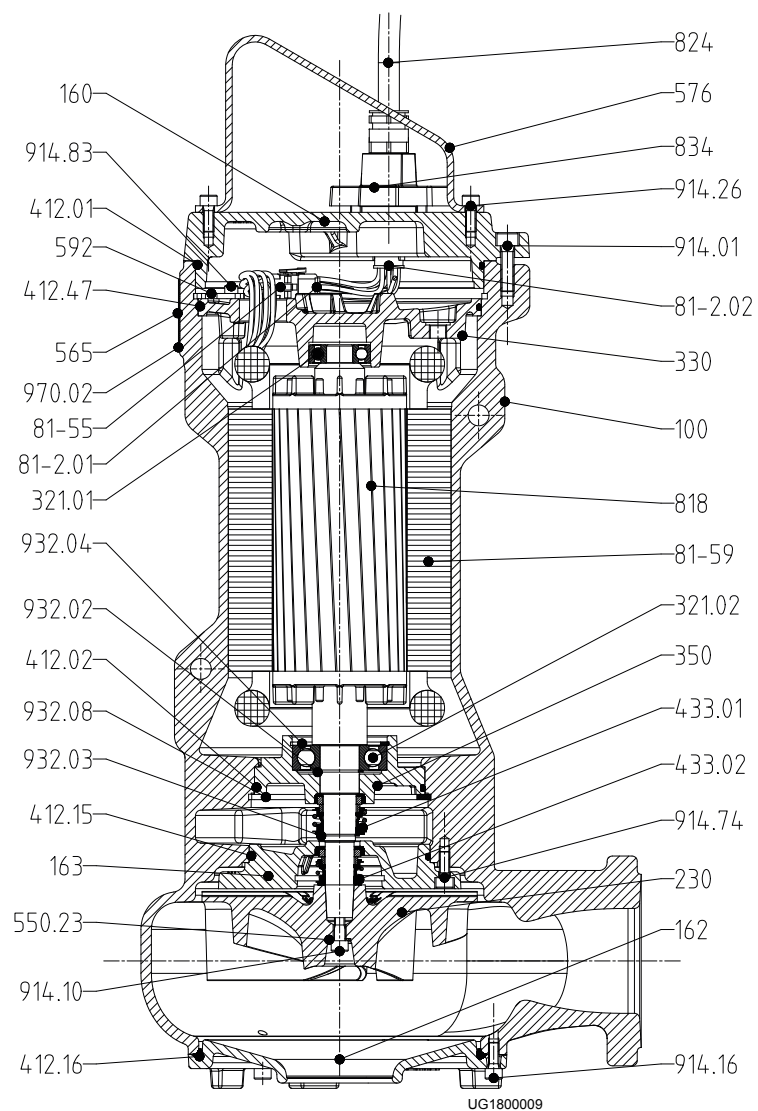


Fig. 14: General assembly drawing for version YS, impeller type F-max

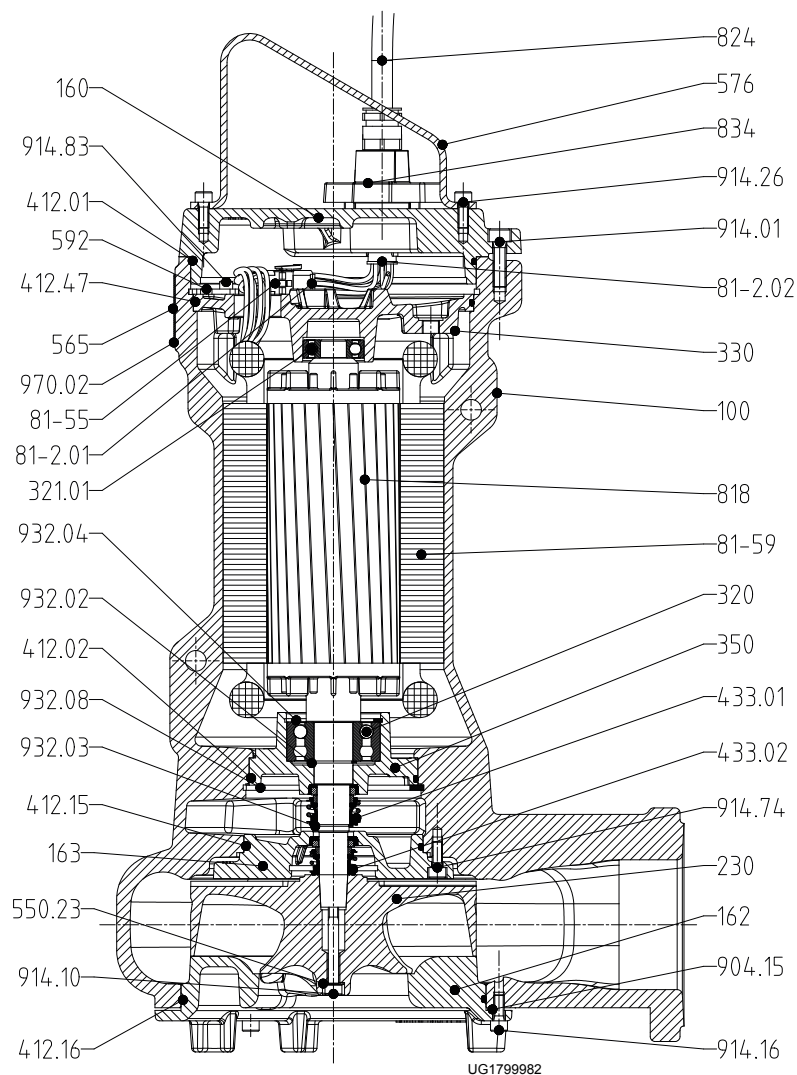


Fig. 15: General assembly drawing for version YS, impeller type D-max

List of components

Part No.	Description	Part No.	Description
100	Casing	576	Handle
160	Cover	592	Plug
162	Suction cover	81-2.01/.02	Plug
163	Discharge cover	81-55	Socket
230	Impeller	81-59	Stator
320 ²⁶⁾	Rolling element bearing	818	Rotor
321.01/.02 ²⁷⁾	Radial ball bearing	824	Cable
330	Bearing bracket	834	Cable gland
350	Bearing housing	904.15 ²⁶⁾	Grub screw
412.01/.02/.15/.16/.47	O-ring	914.01/.10/.16/.26/.74/.83	Hexagon socket head cap screw
433.01/.02	Mechanical seal	932.02/.03/.04/.08	Circclip
550.23	Disc	970.02	Label/plate
565	Rivet		

26) Only used for version with impeller type D-max

27) Only used for version with impeller type F-max

Glossary

Close-coupled design

Motor directly fitted to the pump via a flange or a drive lantern

D-flector

Torque-transmitting element

IE1

Efficiency class to IEC 60034-30: 1 = Standard Efficiency (IE = International Efficiency)

IE2

Efficiency class to IEC 60034-30: 2 = High Efficiency (IE = International Efficiency)

IE3

Efficiency class to IEC 60034-30: 3 = Premium Efficiency (IE = International Efficiency)

Mat. No.

This identification number is composed of an 8-digit numerical code that uniquely identifies a product entered in SAP.



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